User Manual

AI Tools

This manual provides an overview of the Ask AI chat functionality and the Objective AI layer for machine learning, describing their roles, permissions, and integration for advanced analytics.

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CCH® Tagetik
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Chat Ask AI

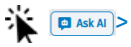
Introduction

Generative artificial intelligence (GenAI) refers to artificial intelligence capable of generating text, images, videos, or other data using generative models in response to requests.

CCH Tagetik, through the **Ask AI chat,** allows you to use Generative Artificial Intelligence across various functional areas of the application.

Ask AI Chat

User | <database applicativo>



Ask AI Chat Window

The Ask AI chat window allows you to enter requests into the text box, either manually or by voice, via the microphone, and send them via the Enter button or keyboard Enter.

Actions on this page

Element	Description
	Ask AI chat **open button** . Clicking on the Ask AI button opens the chat named **Ask AI** ***IMPORTANT**: only visible if function **0004257 - ASK AI into intelligent analytics dashboard** is enabled (only for system roles **$ALL,$AUDITand$USER**).*
	Ask AI chat close button . Clicking on the button closes the Ask AI chat
	Ask AI chat **close button**
	**Send button e**nabled for **send**ing the request entered in the text box. It is only enabled when there is text in the text box
	**Send button d**isabled. It is disabled when: • there is no text in the text box • the user has sent a request and is waiting • a request is being registered
	**Microphone before recording** a request. When green, it is enabled and you can click on it to register a request. When you click on the still green microphone it turns red and bigger.
	**Microphone disabled**. When the Ask AI chat is opened or accessed again after about 10 minutes, the microphone appears disabled for a few seconds. It also appears disabled when the user has sent a request and is on hold.

Element	Description
	**Microphone** for **recording** a **request**. When the microphone is red and bigger, it is possible to record a request that will be written in the text box. During recording, the Enter button is disabled. When the voice stops recording, writing in the text box stops too. It is also possible to stop writing in the text box while a request is being recorded by clicking on the microphone when it is red and larger. When the recording and then the writing of the request stop, the microphone turns green, the icon shrinks and the Enter button is enabled again.
	**Button to open the Languages element** for microphone language selection The button is displayed to the right of the microphone. When clicked, it opens a panel where you can select the language to use for recording your voice command.
	Button for opening the language element disabled. When the Ask AI chat is opened or accessed again after about 10 minutes, the button for opening the language element appears disabled for a few seconds. It also appears disabled when the user has sent a request and is on hold.

Working with Ask AI Chat

Open Ask AI chat

User | <database applicativo>



1. Click on 

2. When the Ask AI chat opens, the system displays a **message** prompting the user to enter a request and indicating the **language** in which a request can be recorded using voice command. The current day and time are shown above the message.

***IMPORTANT**: date and time refer to the calendar year and not the fiscal year.*

Making a textual request in the Ask AI chat

1. Click on 

2. When opening the Ask chat, enter arequest in the chat text box
3. click the 'Enter' button or click 'enter' from the keyboard

   Note: three moving dots are shown while waiting for a response. For each request message and each reply message, the sending and receiving time are shown below respectively

Making a request in Ask AI chat with voice command

1. Click on 

2. When the Ask chat opens, click on the newly enabled microphone button: when the microphone button has turned red and larger, and the Enter button is disabled, make a voice request: the request is recorded and written in the text box
3. Stop writing: when you stop speaking, writing in the text box stops and the microphone and Enter button are enabled again. To stop writing the request, you can also click on the red microphone while recording

Close Ask AI Chat

To close the Ask AI chat window, click on the **X button** at the top right of the chat window or the 

button.

Select the microphone's language

1. Click on 

2. In the Ask AI chat, click the button to open the language selector : a panel opens within the Ask AI chat containing a list of languages to choose from for recording a request via voice command When the language selector opens for the first time, the language selected by default corresponds to the user's application language.
3. Filter the language: in the language selector, type the language you want to select in the search text field, or just a few characters of it. The list of available languages is filtered based on the search.
4. Select language: click on the language of your choice for voice command recording. When the language is selected, the language selector closes.
5. Display the selected language: by hovering the mouse over the open button of the language selector, a tooltip shows the selected language.

***IMPORTANT**: The language selected in the language selector for recording a request via voice command in the Ask AI chat remains stored for each user session. At the start of each new session, the default language (the user's application language) is set.*

History of Ask AI chat

All request and/or reply messages displayed in the chat are stored and entered into the Ask AI and user chat history.

The history of Ask AI chat messages is deleted at the end of each user session.

For each request message and for each reply message in the chat history, the time it was sent and received is shown.

***IMPORTANT**: the history of chat messages is maintained in the different sections of the CCH Tagetik application where Ask AI chat is used, except for Intelligence Analytics which has its own history.*

Error handling in Ask AI

When submitting a request via Ask AI, errors may occur preventing the request from being completed.

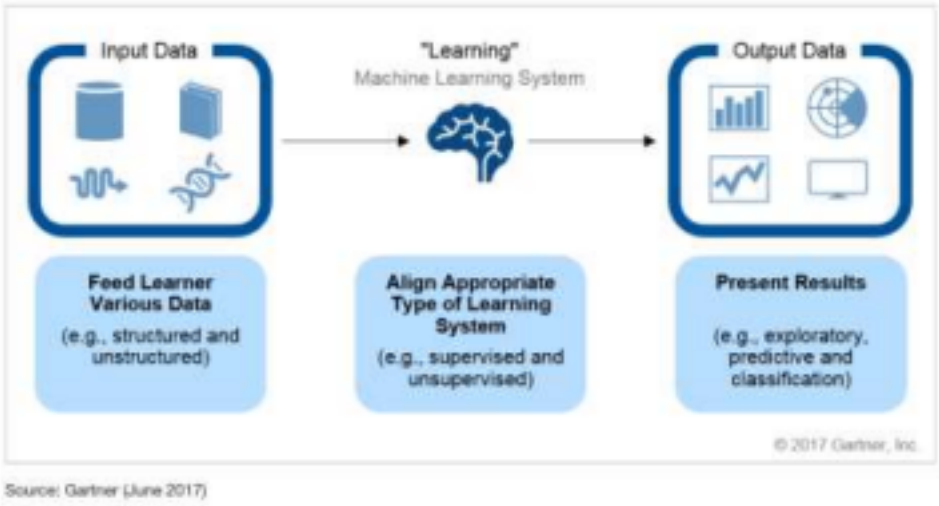
Error types and associated messages

In the event of an error, one of the following messages is displayed in the chat:

Error type	Associated message
Generic error	"A problem appears to have occurred. A problem appears to have occurred. Please try again later or contact support for additional assistance."
Service unavailable	"AskAi not available. Please try again later."
Session expired	Your session has expired due to inactivity. Please reload the page."

Machine Learning in CCH® Tagetik

The Machine Learning can process either structured or unstructured data autonomously (unsupervised) or aided (supervised), for the purpose of producing useful results for the detection of patterns explaining past or future events. The Machine Learning is the ultimate innovation in the Finance industry and, as such, it offers outstanding opportunities.



Source: Gartner (June 2017)

In CCH Tagetik you can enrich the information in the Analytical Workspace thanks to the execution of supervised and unsupervised models. It allows you to train the models with your own data in the Analytical Workspace, analyse the output of the training and also carry out predictions.

The available models are contained in a models library .

Inside the Analytical Workspace you can configure both the Advanced Analytics for the training phase and the Advanced Analytics for the execution phase of the trained model. This allows the users to carry out the prediction from the workflow.

Note: currently, this solution can be enabled only on the Cloud service provided by CCH Tagetik.

Install a model

To enable the installation of an ML model, the admin user must log into the Analytical Workspace.
Here you will find a link in the top right-hand corner called 'Advanced Analytics Models library'.
To access this library, simply click on the **Advanced Analytics Models library** link.
A page pops up showing the list of all available models.
On the left side, you'll find the list, with a (plus) button ✚at the bottom to add a model. In the central area, you can see the model's description and version.

The following is indicated for each model: the code, whether or not it has the training step, and the status of the model.

The model status can be READY, PENDING, DELETING, DEPLOYED, FAILED, DEPLOY_FAILED, DELETED and OFFLINE.

If, when opening the model library, a model is no longer present in the database, it is shown with the status NOT FOUND.

ATTENTION: The PENDING status is displayed in the red box but does not mean that the model cannot be used for data processing. A newly loaded model has its status set to PENDING and then switches to READY.

To add a model, the admin user must click the '+' button. A popup will then appear, containing a field for the code of the model and one for its description.

To install a new model, click the combobox and all templates that can be installed will be shown. Once the user selects a model, the 'OK' button is enabled. When you click OK, the system installs the model.

 If the model is used in an AA, it cannot be deleted.

The ID of the uploaded model can be displayed by clicking on ⋮ next to the desired model.

When the admin selects the Edit option, a window appears displaying the model ID.

 Only the field containing the description of the model is editable.

Model update

The admin can update the model their library with a new version.

To update a model, the following actions must be performed:

1. click on ⋮ of the model
2. click on **Upgrade**. The system displays a **warning**: when clicking **OK** to confirm, the user may lose the training performed on the model and thus may need to repeat the operation with the updated model.

After selecting the model of interest, its version can be displayed in the central screen under the **Model Version** section.

***IMPORTANT**: In case a later version exists, after clicking **OK** the system starts the update. In case there is **no later version** than the one currently in the library, the system displays the message: 'There are no new versions for this model'.*

If you refresh the page you will see the model in the **Upgrading** status.

When the page update operation is complete, the model will be in the **Ready** status and the updated version will be visible in **Model Version**.

***Note**: If the operation is unsuccessful, the model remains Ready and at the version before the operation and in the Model Version paragraph the system indicates that the upgrade was unsuccessful.*

Locking and unlocking user models

The admin can lock and unlock one or more models in their library.

To **lock** a model, the following actions must be performed:

1. select the model and click on ⋮
2. click on **Lock Model** The system shows a warning: by clicking on **OK**, the user will no longer be able to launch Train and Execute processes related to that model and the model will be in the **BLOCKED_BY_USER** status

   Should the user try to launch a data processing related to this model, the data processing will not be executed, because the model's status is different from **ready**.

   To **unlock** a model, the following actions must be performed:

   1. select the locked model and click on ⋮
   2. click **Unlock Model**.

   ***Note**: This feature is only available for models that are in the **Ready** status.*

Lock from CCH TagetikTagetik

Models can also be locked from CCH Tagetik. In this case, the models will be shown in the **BLOCKED_BY_TAGETIK** status.

Should the user try to launch a data processing related to this model, the data processing will not be executed, because the model's status is different from **ready**.

IMPORTANT: *The user cannot unlock models that are in the* **BLOCKED_BY_TAGETIK** status.

Analytic Workspace - Advanced Analytics

An Advanced Analytics (AA) represents a set of activities that have to be run for a Machine Learning (ML) algorithm defined by a model contained in the models library.

An AA can belong to one Analytical Workspace (AW) only but several AA can be defined for each AW.

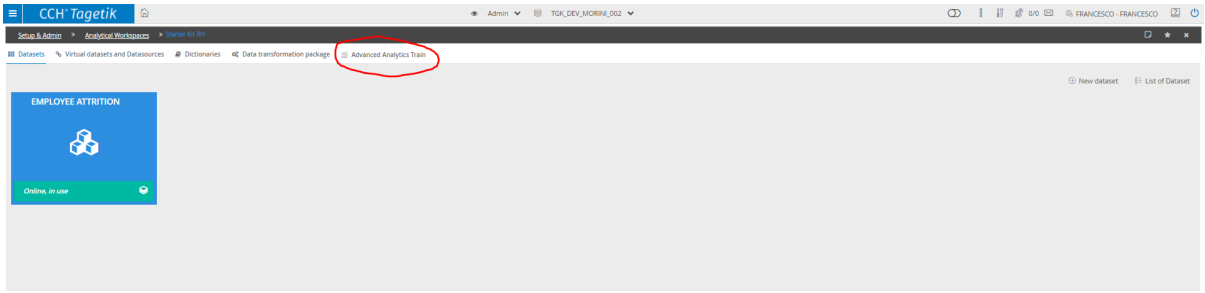
An AA can be either of "Train" type or of "Execute" type; thus, you'll have two different objects:

- Train Advanced Analytics
- Execute Advanced Analytics

A Train Advanced Analytics is the object containing the information about the ML data processing for the execution of the training of a model contained in the models library. Whereas, an Execute Advanced Analytics is the object that stores the information about the predict phase or the execute phase of a model.

Train Advanced Analytics

To reach the Train Advanced Analytics, it is necessary to open the Analytics Workspace and then click on an AW. The page of the selected AW pops up. In this page, the admin user will have to click on the "Train Advanced Analytics" tab to reach the Train Advanced Analytics.

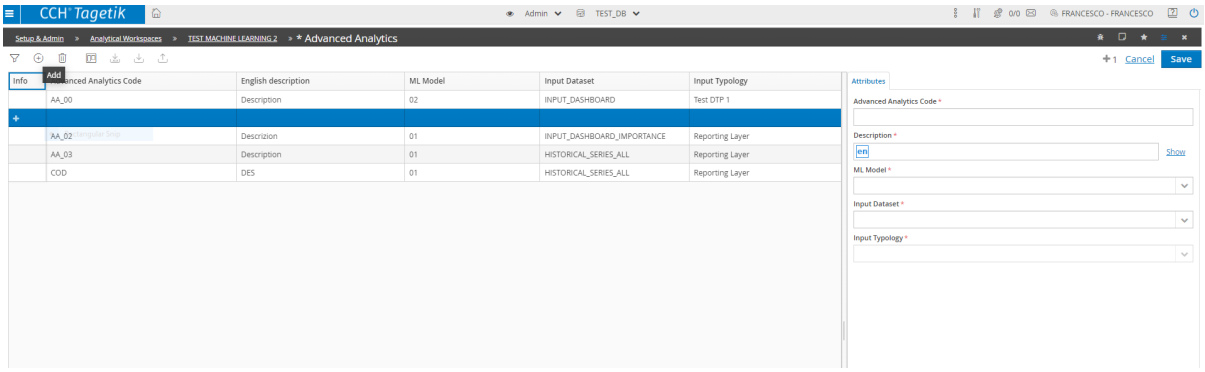


To create an AA of Train type, just click on the button  in the toolbar at the top of the window. A new row will be added in the central pane, and the right panel will show all the fields to be set up. Those fields are:

- **Advanced Analytics Code**: unique code that identifies an AA; this field is mandatory.
- **Description**: description of the AA of Train type; this field is mandatory.
- **ML Model**: here the user can choose the model, included in the models library, with which the Machine Learning data processing will be run; this field is mandatory.
- **Input Dataset**: allows the user to choose which dataset will be used as a source of data that will be then used for the ML data processing; this field is mandatory.
- **Input Typology**: specifies the typology that will be used as input; this field is mandatory.

After setting all the fields up, to co complete the operation the user has to click on 

. The new AA is now created.

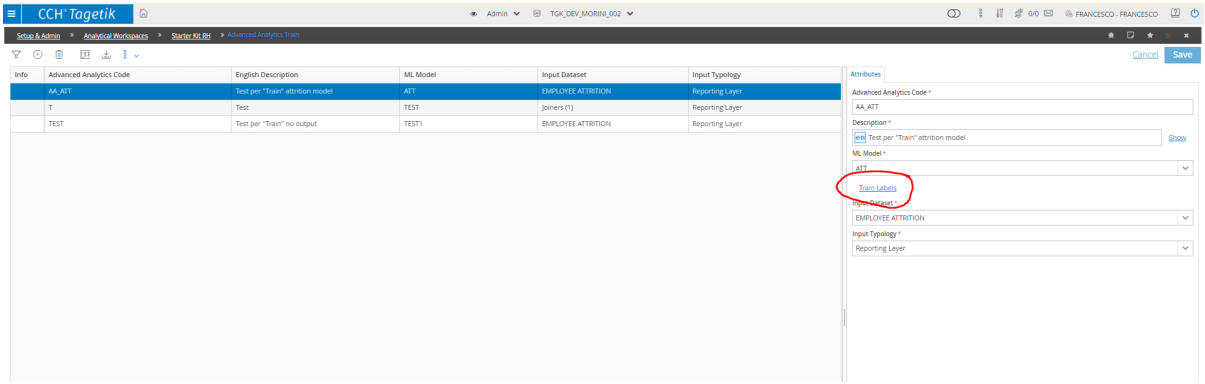


Any new AA can be either modified or deleted. To delete an AA, just select it from the list and click on "Delete"  , then click "Save"  .The following warning message appears: "This advanced analytics may be in use, do you still want to delete it?". This message appears because if the AA is in use in a Workflow task, it will be removed from that task too. If you still want to delete it, just click Yes and the AA will be removed.

To modify an AA, as with the deletion, just click on the AA you want to modify and all the information about the AA will be displayed on the right hand side; every piece of information can be modified. Once the desired fields have been modified, just click on the "Save" button  for the changes to be saved.

Train label

Once the admin user has created and saved an Advanced Analytics of Train type, the "Train Labels" link will appear below the "ML Model" field.

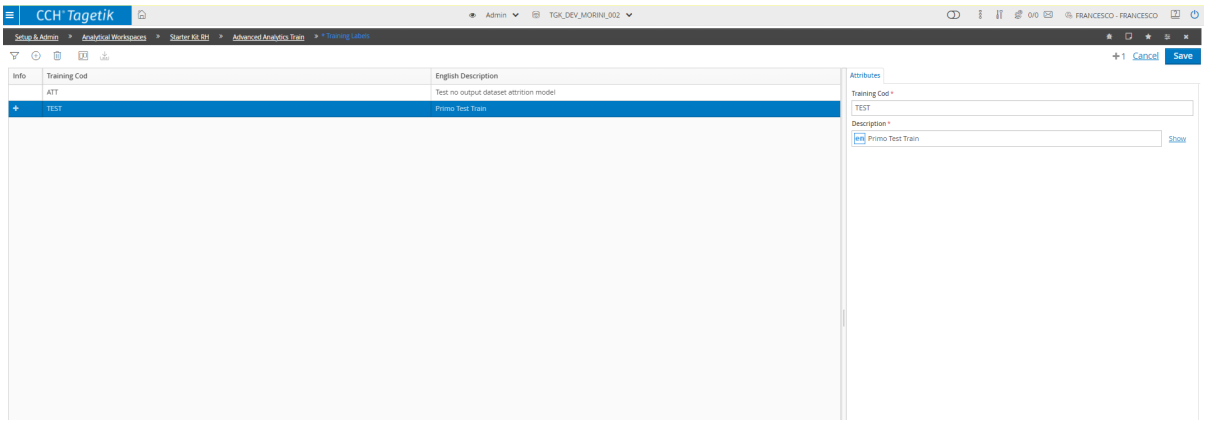


Here it is possible to save some labels for the trainings that can be used in the ML data processing. Indeed, once the ML data processing has started, it will save the train Model after the inference. The model will be serialized and saved. The Train Label represents the link to the trained model.

To create a Label, just click on the button  in the toolbar at the top of the window. A new row will be added in the central pane, and the right panel will show all the fields to be set up. Those fields are:

- **Training Code**: unique code of the Train Label; this field is mandatory.
- **Description**: description of the AA of Train type; this field is mandatory.

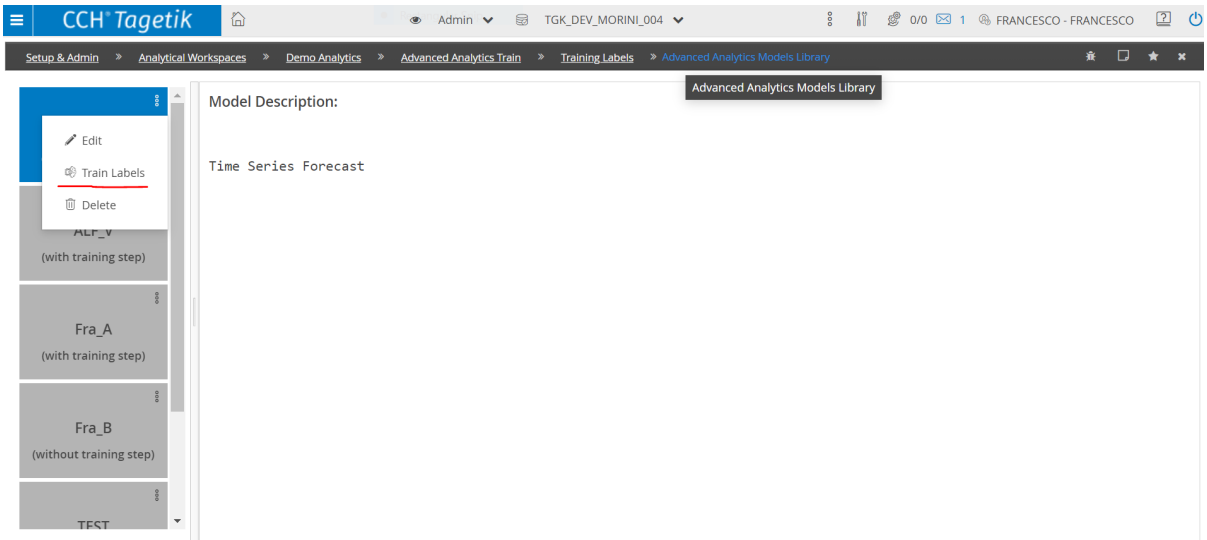
After setting all the fields up, to co complete the operation the user has to click on  . The new AA is now created.



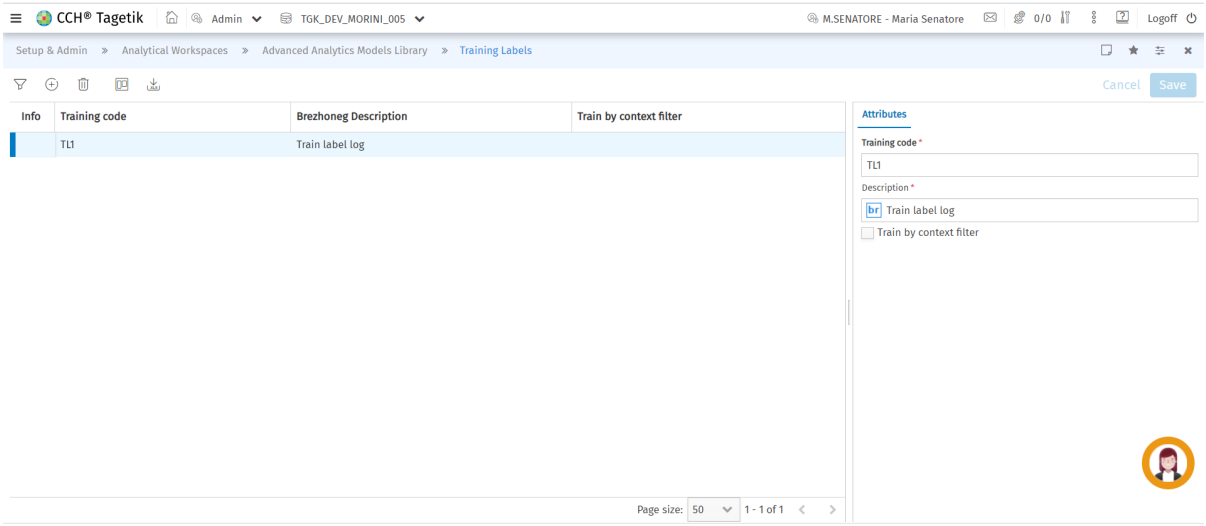
Once at least one Train Label has been created, it can be modified or deleted. To delete a previously created Train Label, simply select the one to be deleted from the list, and click on the "delete" button , and click on the button Save.

To edit a Train Label, as in the case of deletion, simply click on the desired Train Label, on the right-hand side all information concerning the Train Label will be loaded; it will all be editable, in fact the user (Admin) can edit all the fields of their choice, after completion simply click on the Save button and the changes will be saved.

The entry point of the Training Label page can also be accessed by going to the template library view, in fact, selecting one of the templates already loaded, press the "action menu" button and select the "Train Labels" link.



This will open the link to the Train Labels page for the chosen model.



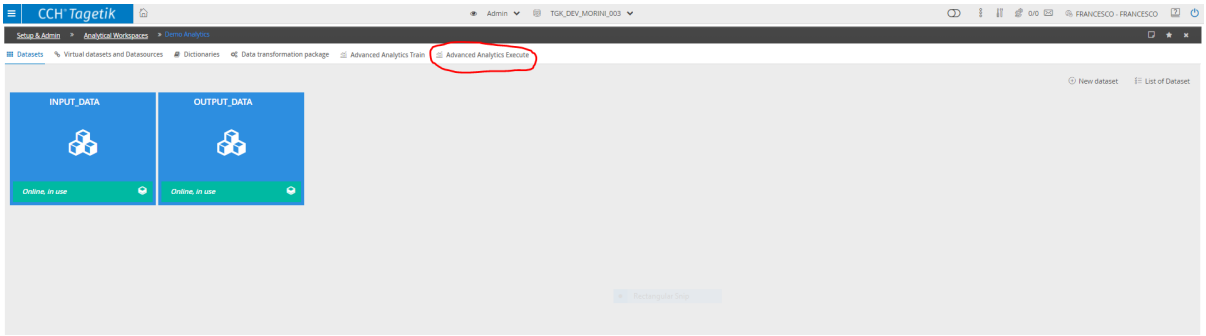
The data processing log file can be downloaded from Audit after selecting the data processing.

The downloaded .zip file, whose name follows the structure *ML_LOG_CORRELATIONOID_TIMESTAMP* (e.g. ML_LOG_2C9480F47E659B7B017E670D2E8B4FEE_1642499276728), contains one or more text files with the following format *ELAB_ElaborationID.log* (e.g. ELAB_fa6f314b-3a66-418f-aa52-0909807ca41d.log), one for each Train or data Execute processing performed.

The log files contain information on the outcome of the data processing that the Data Scientist, author of the model, considers important to report.

Execute Advanced Analytics

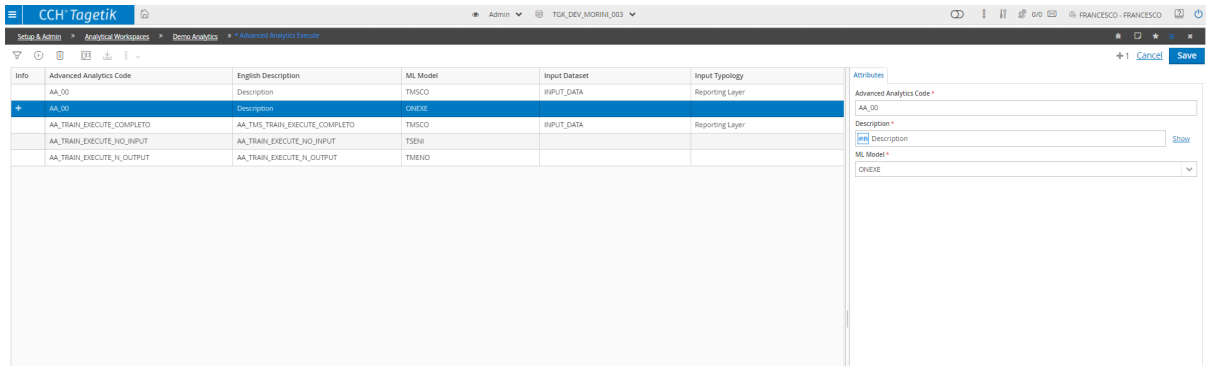
To reach the Execute Advanced Analytics, it is necessary to open the Analytics Workspace and then click on an AW. The page of the selected AW pops up. In this page, the admin user will have to click on the "Execute Advanced Analytics" tab to reach the Execute Advanced Analytics.

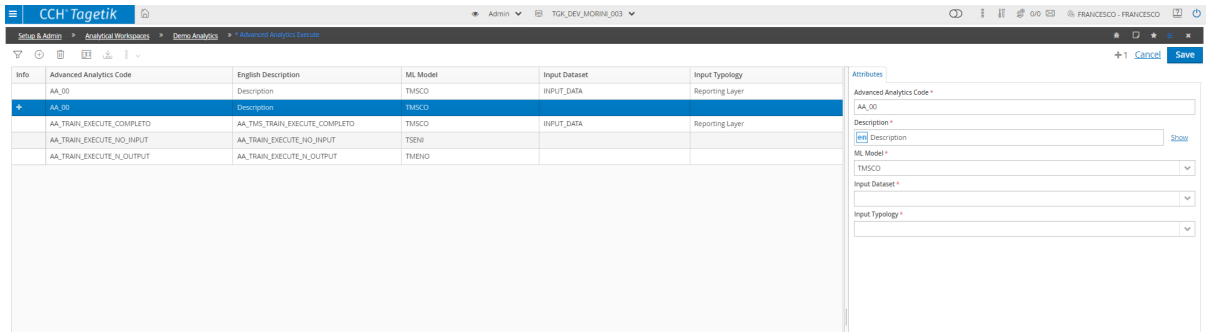


To create an AA of Execute type, just click on the button ⊕ in the toolbar at the top of the window. A new row will be added in the central pane, whereas the fields to be filled in will be displayed in the right hand panel. The last two fields, **Input Dataset** and **Input Typology,** are displayed only if they are set up in the Model uploaded in the "models library" area in the "descriptor.json" gli "inferenceParameters" file. The fields to set up are:

- **Advanced Analytics Code**: unique code that identifies an AA; this field is mandatory.
- **Description**: description of the AA of Train type; this field is mandatory.
- **ML Model**: here the user can choose the model, included in the models library, with which the Machine Learning data processing will be run; this field is mandatory.
- **Input Dataset**: allows the user to choose which dataset will be used as a source of data that will be then used for the ML inference; this field is mandatory.
- **Input Typology**: specifies the typology that will be used as input; this field is mandatory.

After setting all the fields up, to co complete the operation the user has to click on Save. The new AA is now created.





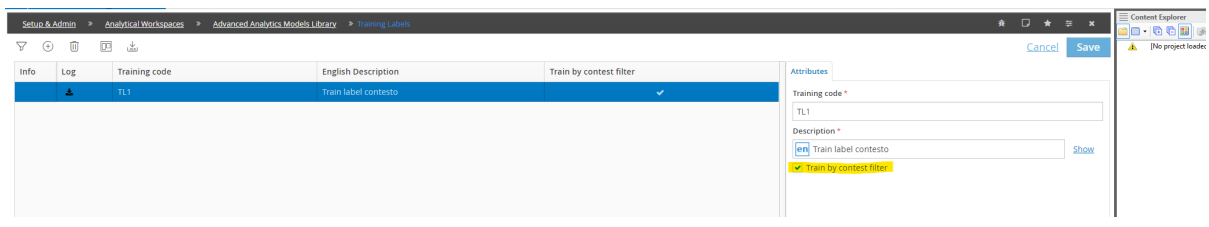
Any new AA can be either modified or deleted, just like any AA of Train type. To delete an AA, just select it from the list and click on "Delete" , then click "Save" Save .The following warning message appears: "This advanced analytics may be in use, do you still want to delete it?". This message appears because if the AA is in use in a Workflow task, it will be removed from that task too. If you still want to delete it, just click Yes and the AA will be removed.

To modify an AA, as with the deletion, just click on the AA you want to modify and all the information about the AA will be displayed on the right hand side; every piece of information can be modified. Once the desired fields have been modified, just click on the "Save" button Save for the changes to be saved.

Train and Execute by context filter

This function allows you to have model train that depend on the launch filter.

In order to define whether a label is to be saved by launch filter, go to the Train label page and select the check: 'Train by context filter'.



When launching a train configured in this way, all information relating to that launch (process, scenario, period, Entity/analytical dimension) will be stored.

⚠ If a train is launched with the same filter combination, the information will be overwritten and associated with a new training id.

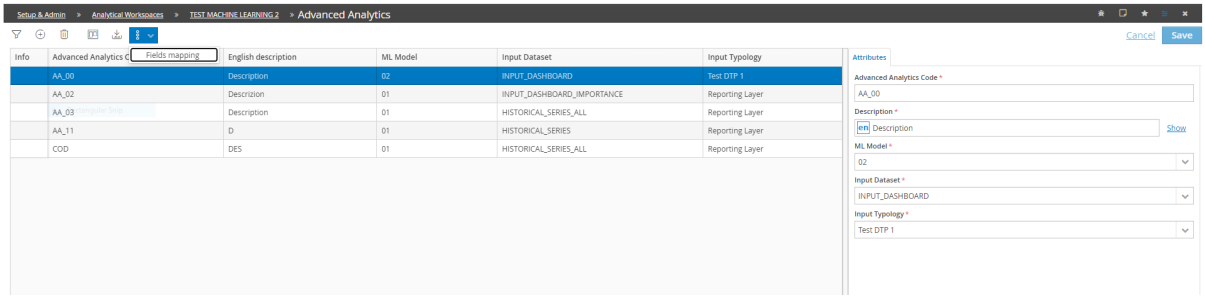
If an execute is launched with the same filter combination as the train, the data processing will end successfully. Conversely, if an execute is launched with a launch filter to which a train does not correspond, the data processing fails by displaying an error message: 'No train exists with the same launch context'.

⚠ When the 'Train by context filter' flag is set on an existing train label, i.e. you want to make the train label by context filter, a warning message is displayed warning that the data on the train label will be lost. Similarly, if an existing train has its flag removed, the same warning message will be displayed and data will be deleted.

Advanced Analytics - Fields mappings

This option allows the user to realise a mapping of the fields; it allows linking to a selected input dataset the fields that will have to correspond to those present in a file named "descriptor.json" contained in the model. These fields represent the data that will be sent to the Machine Learning engine by the data processing. Here the user can select the output dataset in which the values of the ML inference will be stored.

To add the section about the "Fields Mapping", the user has to click on the three blue dots ⋮ in the Advanced Analytics page and then click on "Fields Mappings". A new page will pop up.



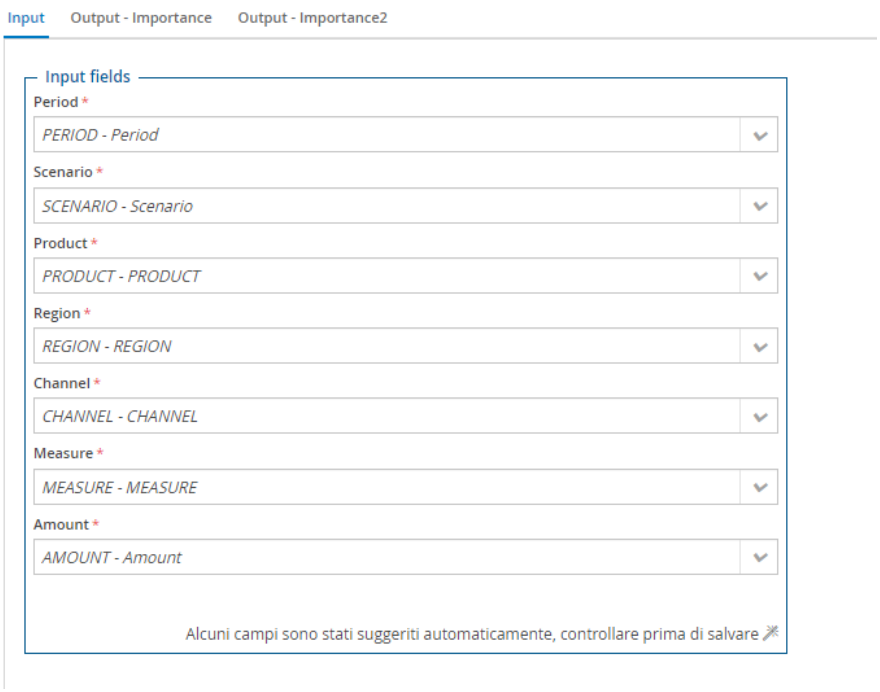
The fields mapping page displays: at the top a set of tabs, an "Input" tab and one or more "Output" tabs according to the number of outputs in the "descriptor.json": the "Outputs" are displayed in the same order as in the "descriptor.json" and each one is named according to the "OUTPUT - *descrizione*" standard, where the description corresponds to the "*descrizione*" field of the json. In the middle of the page there are two boxes, one for the fields mapping and one for the parameters. These boxes represent the fields of the dataset and are used as data to perform the inference. These fields are the same as those in the "descriptor.json" file contained in the model uploaded in the models library.

Changes made to the Fields mapping are saved using the appropriate Save button, which will save them in memory and result in exiting the window and, at the same time, returning to the AA page.

Loading the "*Fields mappings*" page enables an autofill function, both for the "Input" tab and for the "Output" ones, that fills the parameters fields with possible mappings based on the equivalence between the names and the parameter types of the output and those of the selected dataset.

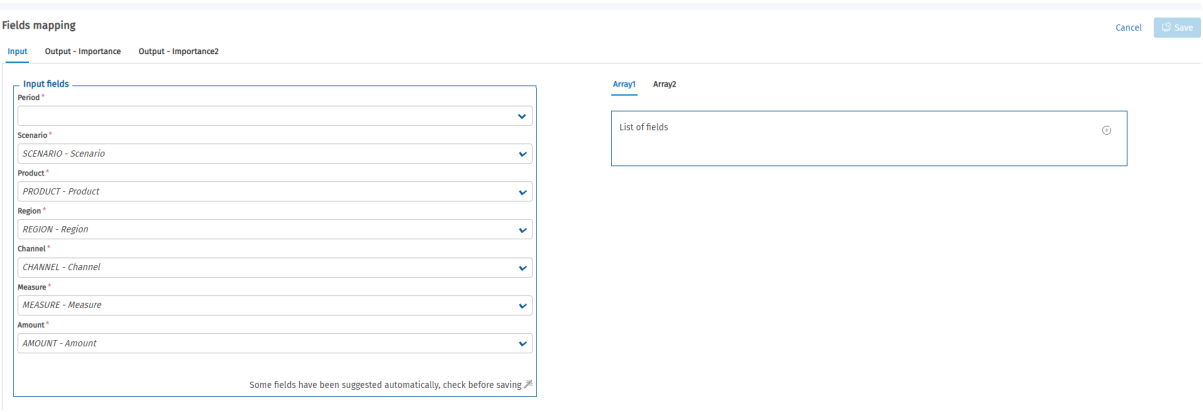
In the case of automatic mapping, suggested parameter values are shown in italics, and a message in the bottom right-hand corner of the fields box is displayed to warn the user that some fields have been automatically suggested and should be checked before saving.

Once saved, if you return to the Fields Mapping page of the same AA, the message is no longer displayed.

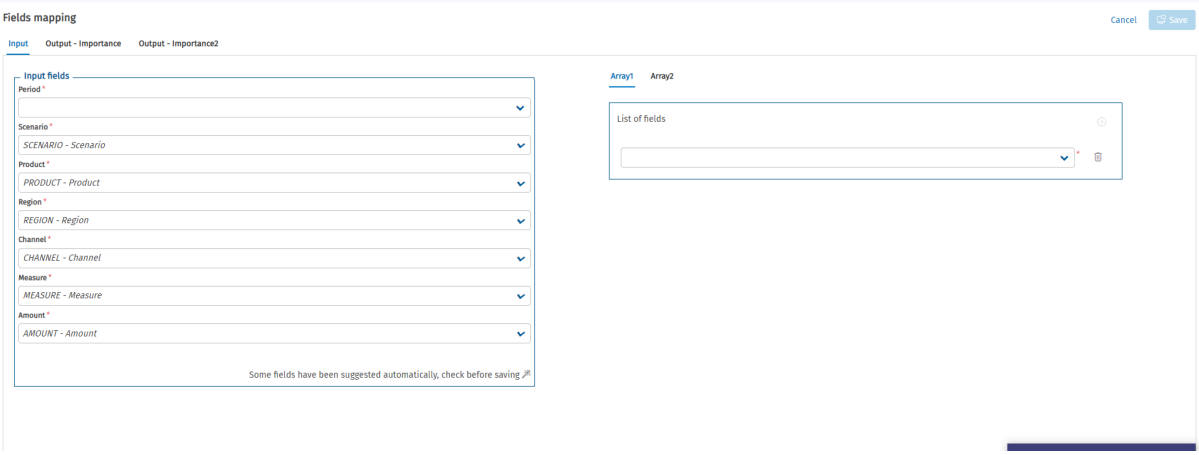


In case the selected model has one or more array elements, both for Advanced Analytics Train and Advanced Analytics Execute, additional boxes will be loaded along with the *typical* field box, for both the Input and Output tab(s), referring to the array elements subdivided into different tabs, where the user can map the fields that will make up the array.

By default, the array element does not show any comboboxes to be filled with a field, but only the different Array tabs. The combobox to be filled, at the first Fields Mapping in the Advanced Analytics configuration, must be added.



For each array element, clicking on the ⊕ button adds a combobox in which to select the field that will be part of that array: this combobox can then be deleted from the array element by clicking on the delete button 🗑 linked to each combobox.



Each array element, for both the Input and Output tab(s), must contain at least one field, in order for the *Fields Mapping* page to be saved.

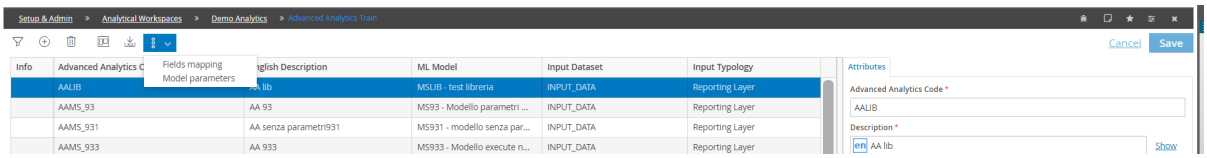
The fields of each array of the Input tab or of the Output tab/s cannot be equal to one of the typical Input or Output fields or to the field of another array in the same tab.

If you enter equal fields for two or more arrays or an array with *typical* Input or Output fields, and try to save, an error message will notify you that the same information cannot be associated twice, and the saving of the *Fields Mapping* page does not take place.

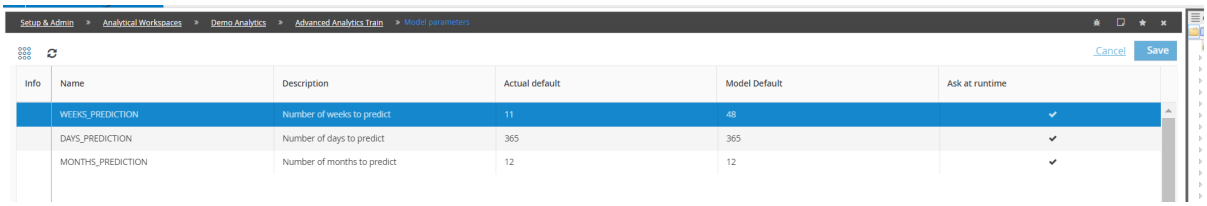
Advanced Analytics - Model parameters

This section allows the users to display and edit the model parameters.

To reach the 'Model parameters' section, a user who is already on the Advanced Analytics page can simply click on the three blue dots ⋮ and then click on 'Model parameters'. A new page will pop up.



The new page will display the list of parameters contained in the descriptor.json of the model with which advanced analytics is associated.

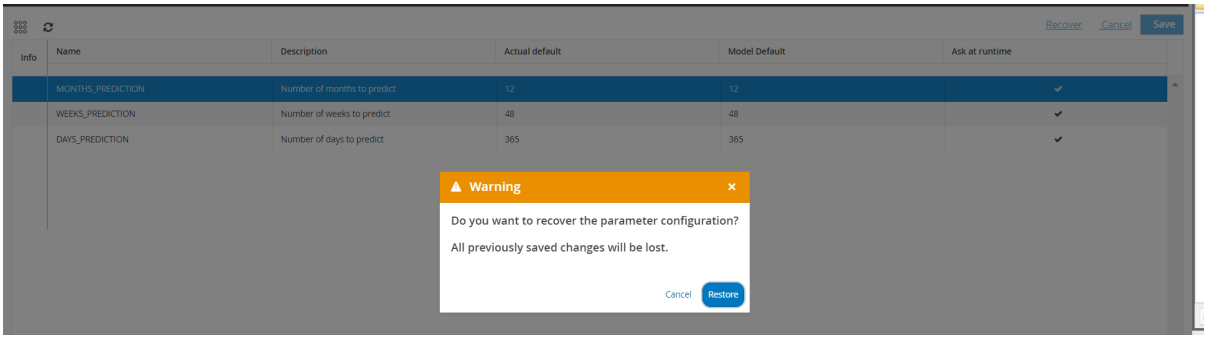


For each parameter, the name, description, current default value and the value contained in the model's descriptor.json and the ask at runtime are indicated.

By default the ask at runtime is set to true and only the fields 'current default' and 'ask at runtime' are editable.

If Ask at runtime is set to false, the current default value is mandatory.

The user who changes the values contained in the 'current default' field and clicks on Save can retrieve the default values contained in the descriptor.json by clicking on the 'Reset' button in the top right-hand corner.



In this case, a message is displayed: 'Do you want to reset the parameter configuration? All previously saved changes will be lost'. When the user clicks on Restore, the current default value is set to the default value of the model with the 'ask at runtime' fields set to true for all parameters; if the user clicks on Cancel, no changes are made to the data and the warning message is closed.

If an advanced analytics is associated with a model where no parameters are defined in the descriptor.json, a message will be displayed: 'The linked ML model hasn't input parameters'.

If a user defines advanced analytics and does not access the model parameters, all parameters will be displayed with the default values contained in the descriptor.json when the task is launched.

If a user accesses the model parameters, only parameters with 'ask at runtime' to true and with the default value equal to the value set in the 'current default' field will be displayed when the task is launched.

WARNING: When editing the model associated with an advanced analytics, the parameters will be deleted and must be re-entered.

Data processing from Excel

The data processing described above can be set up and run also from Excel.

To be able to set up and run a data processing from excel, first of all it is necessary to configure the Advanced Analytics as described in the Advanced Analytics section, as well as to create a Train Label and the mapping and to upload a model into the models library area.

Entry point

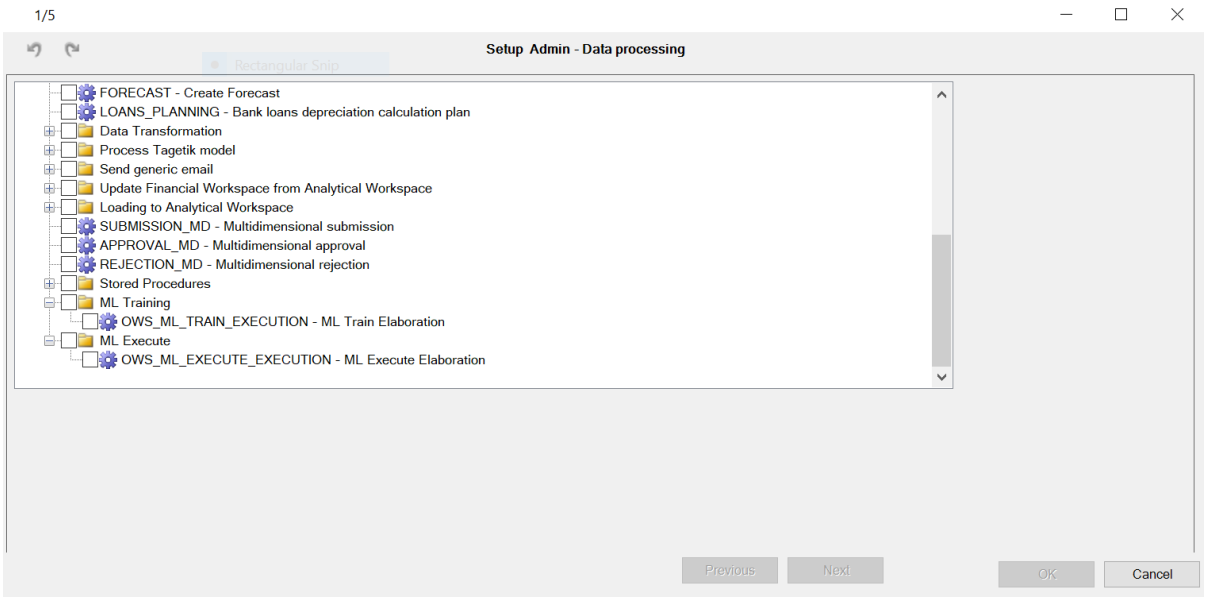
Before creating a ML taskin the ribbon area of CCH Tagetik, the user has to upload a model, as shown in the models library section, then they have to create an Advanced Analytics of Train and/or Execute type, as shown in the "Advanced Analytics" section and finally they have to create the Train Label as shown in the "Train Label" section in Advanced Analytics.

Ml Training

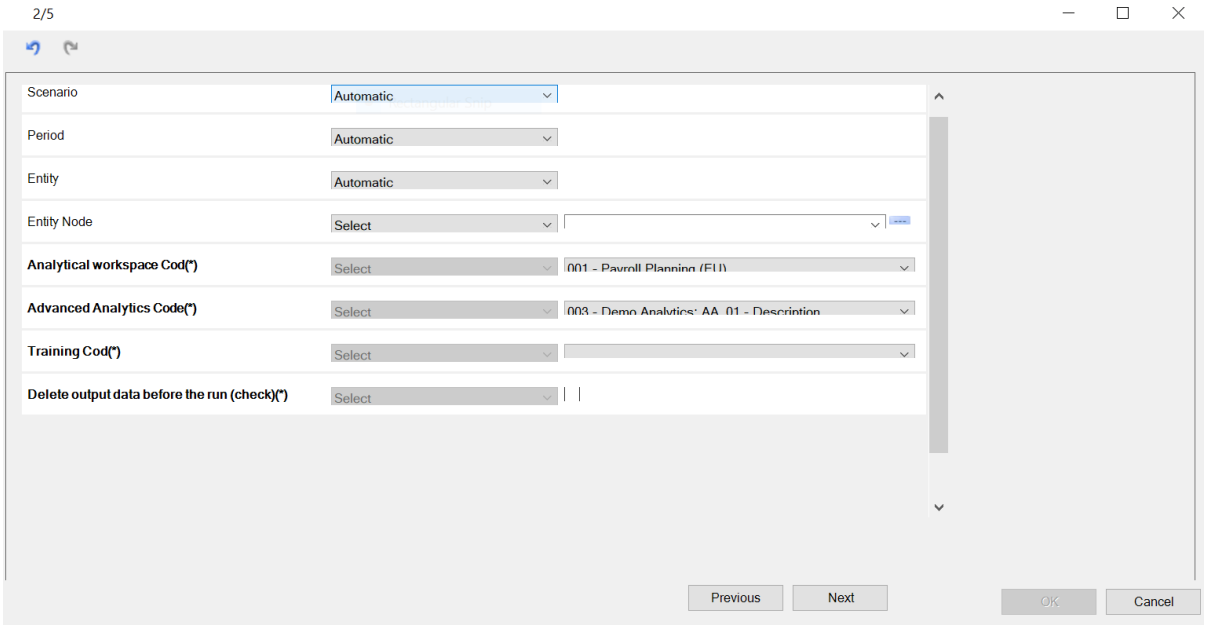
After logging into the proper area of Excel, it is necessary to open a spreadsheet by clicking on the "Design" button, then selecting a db and clicking on the form to open. After that, select the "Data Processing" button in the "Design" area at the top of the page. A pop-up window will appear showing the list of available data processing, in two different folders:

- Ml Training
- MI Execute

containing respectively the Train and then Execute data processing.



To create a Train data processing, it is necessary to select "OWS_ML_TRAIN_EXECUTION" and then click OK. The Data processing configuration window will appear.



Here are all the fields to be set up, like in the Advanced Analytics area.

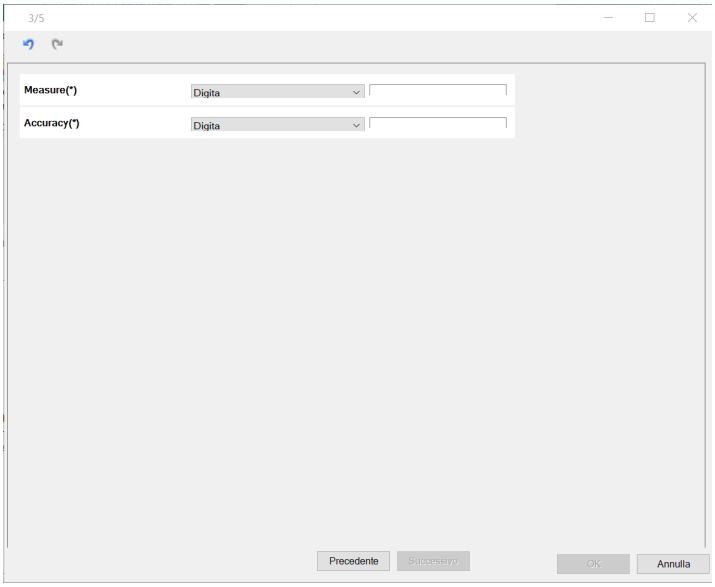
Here are the parameters for the train data processing:

- **Analytical workspace Code**: source AW for input and output data.
- **Advanced Analytics Code**: source Advanced Analytics for input and output parameters. It is identified as: AW - Name Advanced Analytics
- **Training Cod**: Train Label selected in the Train Labels section. It is identified as: Model Name - Train Label. In fact, it refers to the train label created in the relevant section.
- **Delete data before the run (check)**: flag that allows you to delete the data of a previous data processing.

Please, be aware that if you choose a wrong combination of parameters, the data processing will end with error.

Once all fields are set up, click Next.

If some parameters are required in the descriptor.json, a window in which to define them will pop up.

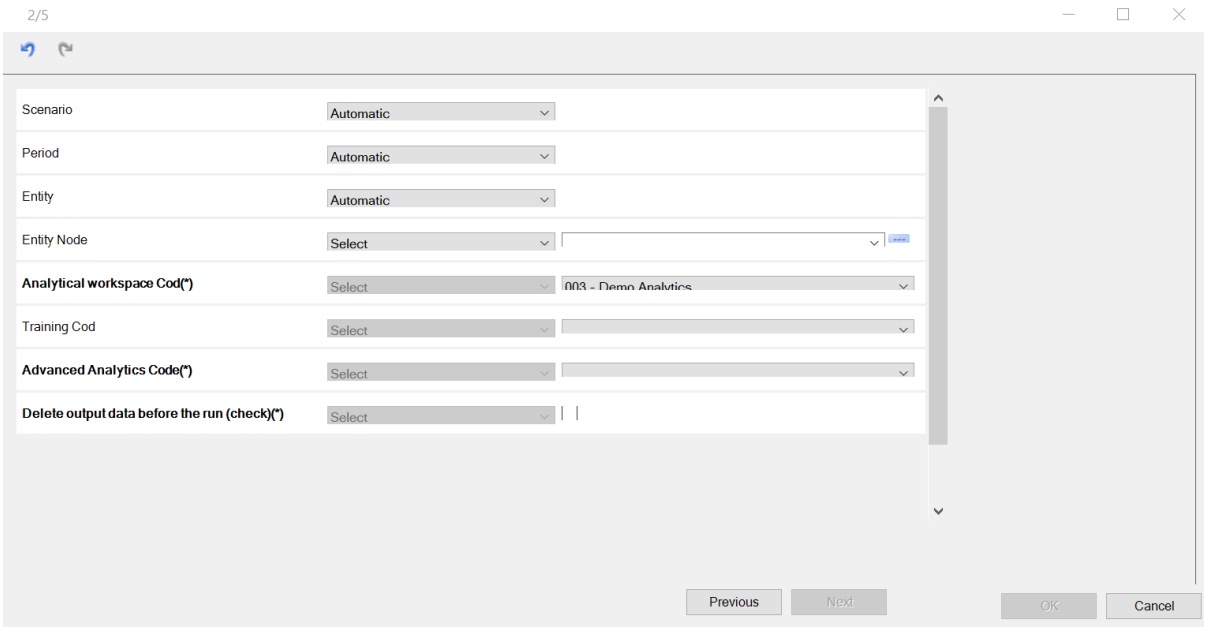


Attention : the managed parameters are string type only.

Click on Next and select the desired options. If you select "At Runtime, show a link for the run", once the configuration is completed, the system will display a link for the execution of the data processing. After that, click on Next, pick a name and a description for the Data Processing, and then click "OK".

MI Execute

To create an Execute data processing, select "OWS_ML_EXECUTE_EXECUTION" and then click OK. The Data processing configuration window will appear.



Here are all the fields to be set up, like in the Advanced Analytics area.

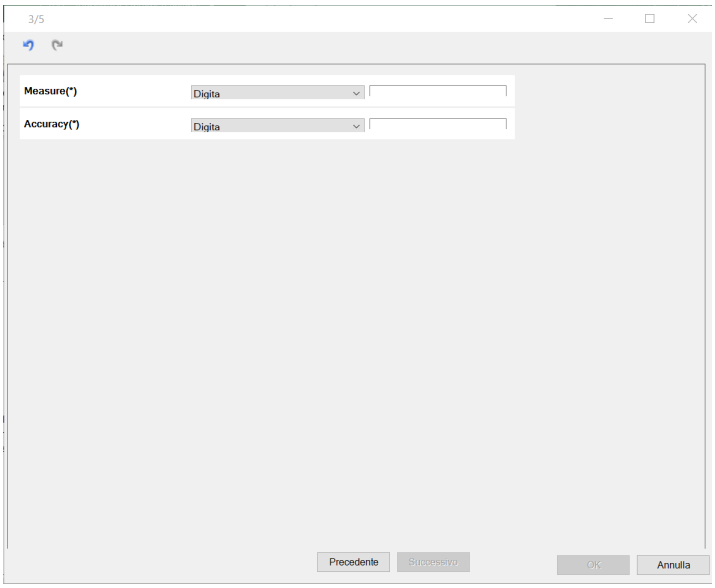
Here are the parameters for the train data processing:

- **Analytical workspace Code**: source AW for input and output data.
- **Training Cod**: Train Label selected in the Train Labels section. It is identified as: Model Name - Train Label. In fact, it refers to the train label created in the relevant section.
- **Advanced Analytics Code**: source Advanced Analytics for input and output parameters. It is identified as: AW - Name Advanced Analytics
- **Delete data before the run (check)**: flag that allows you to delete the data of a previous data processing.

Please, be aware that if you choose a wrong combination of parameters, the data processing will end with error.

Once all fields are set up, click Next.

If some parameters are required in the descriptor.json, a window in which to define them will pop up.



Attention : the managed parameters are string type only.

Click on Next and select the desired options. If you select "At Runtime, show a link for the run", once the configuration is completed, the system will display a link for the execution of the data processing. After that, click on Next, pick a name and a description for the Data Processing, and then click "OK".

Data processing by Job

Machine Learning data processings are also configurable, launchable and scheduled via Job.

In order to be able to configure and directly launch or schedule the launch of a Machine Learning data processing via Job, one must: as a precondition:

- Have uploaded a model as in the Upload of a Install a Model

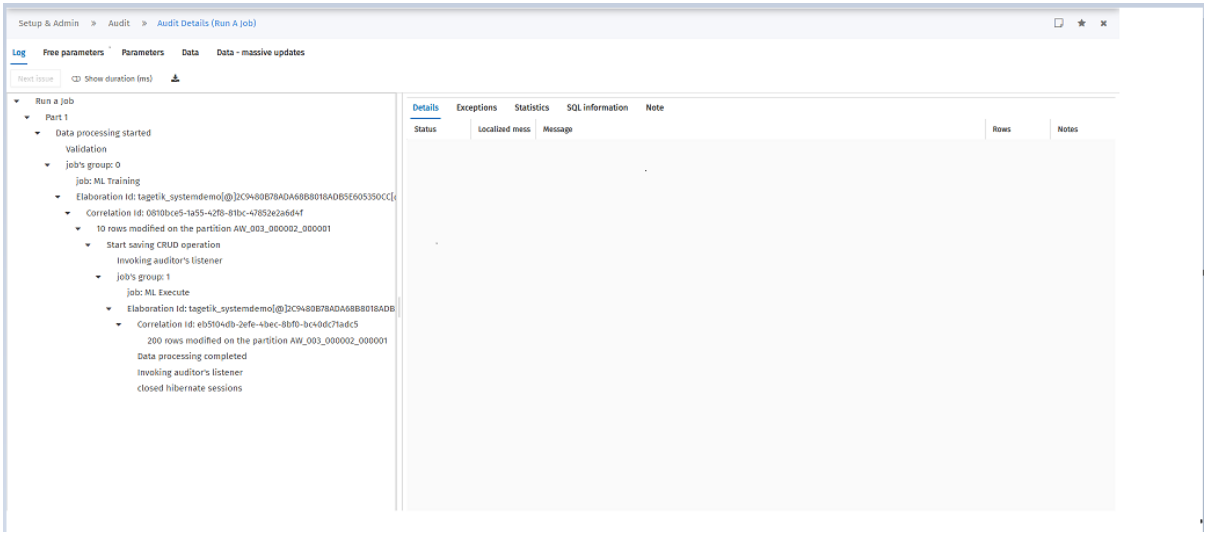
- Have configured Advanced Analytics, Train and/or Execute type, as described in the Machine Learning Advanced Analytics section.

- Finally, one must have created a Train Label and completed the mapping of fields, as highlighted in the sections 'Train Label' and 'Advanced Analytics - Fields mappings' also in Advanced Analytics.

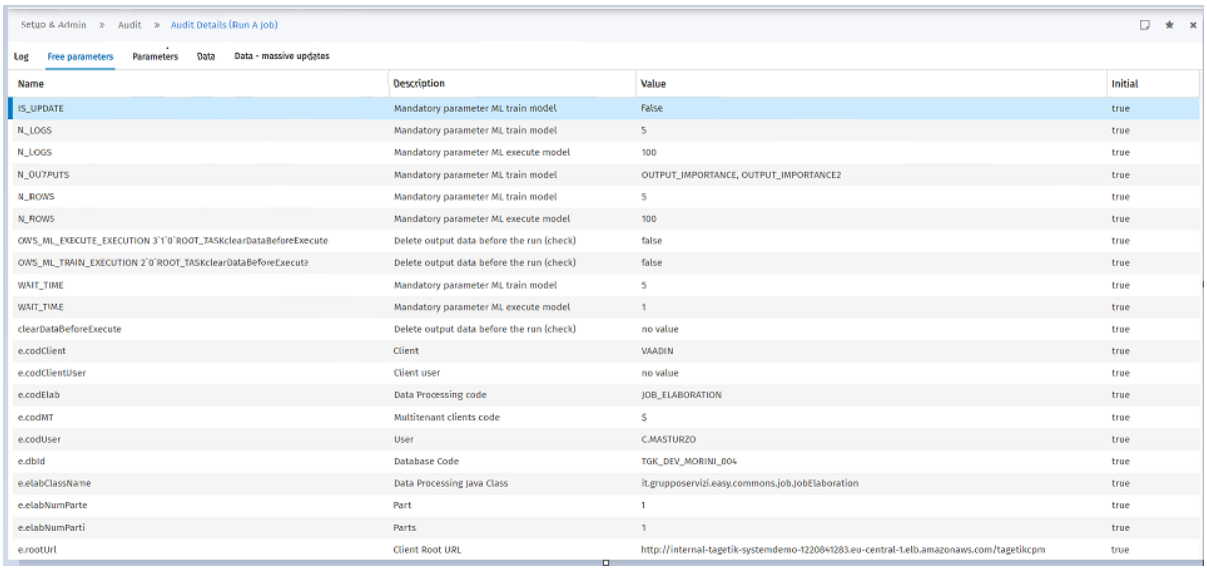
ML Training and ML Execute from Job

After having added an ML Train and/or Execute task to the graph of a previously created Job, as described in the "Define Job" section under Job, and having defined all the necessary parameters(ML Training and ML Execute tasks), you can run or schedule a Job by clicking on the *Review and Launch* button from the read-only view of its graph, and finally click on *Launch* or *Schedule*, depending on the operation you wish to run.

After running the Job for Train and/or Execute data processing or having scheduled it and waited for it to start at the scheduled time and day, you can view the details of the finished Job in the Audit window.



If the Machine Learning model, used in the configuration of Advanced Analytics, selected in the definition of task parameters within the Job, is a model with parameters, it is possible to view the parameters read at runtime by clicking on the 'Free parameters' tab in the Audit details relating to the Job:



- If the parameters of the model are set in Advanced Analytics and the current value of the parameters is not empty, they are correctly displayed in the 'Free parameters' tab in the Audit Job details.

- If the model parameters are set in Advanced Analytics and the current value of the parameters is empty, they are correctly displayed and have 'no value' as Value in the 'Free parameters' tab in the Audit details for the Job.
- If the model parameters are not set in Advanced Analytics (they were never displayed and set in the 'Model parameters' window), the default values contained in the model's descriptor.json are shown in the 'Free parameters' tab in the Audit details for the Job.

 If the values of the model parameters are empty (empty value in the descriptor.json of the model) and are not set in Advanced Analytics (they have never been displayed and set in the 'Model parameters' window), the Job fails and an 'Impossible to retrieve the runtime parameters of the model' error message is displayed after the run. Try adding a default value.

 Warning: If two or more Machine Learning data processings have been added to the same Job and have the same values of one or more parameters of the model, only one row for that parameter or those parameters with the common value is shown in the 'Free parameters' tab in the Audit details for the Job.

ML Workflow Tasks

In this section, we will explain how to:

- Properly set up the workflow tasks and
- run the task and ensure the ML inference occurs.

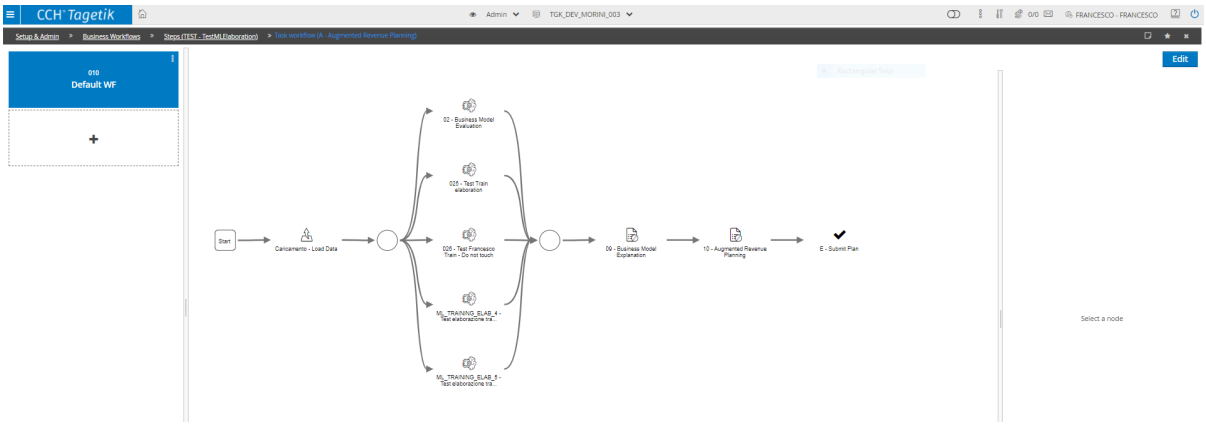
The ML inference tasks ensure that the Dataset input for the ML engine trains a model during the Train phase. The outcome of such phase will be the trained Model whose reference is stored into a Train Label and whose creation is specified in "Train Label", in the Advanced Analytics section. After the Train phase, you can run the Execute task in which the ML engine runs the inference; this phase can take as input a Dataset, the reference to the trained Model and generates a Dataset.

 Based on the Model uploaded into the models library, the ML inference will generate some rows to be inserted into a Dataset. If it produces OID rows, these will be inserted into a Dataset and will update the existing rows; whereas, if the produced rows are without OID, these will be rejected. In some cases, always based on the uploaded Model, the rows produced by the ML inference will be inserted as new rows into a Dataset.

Entry point

Before creating a ML task, the Admin user has to upload a model, as shown in the models library section, then they have to create an Advanced Analytics of Train and/or Execute type, as shown in the "Advanced Analytics" section and finally they have to create the Train Label as shown in the "Train Label" section in Advanced Analytics.

To create a task, the Admin user has to access the Business Workflow area, click on the desired Workflow, click on "Actions" ⋮ and then select "Steps". The user will have to choose and click on one of the steps in the steps tree. A window will pop up showing on the left hand side the Workflow identified by a code and, in the middle, the detail of the workflow displayed in the form of a tree.



Now, the Admin user can select the workflow in which to insert the task for the ML data processing and then click on the "Edit" Edit button in the right hand corner. A bar will appear showing all the tasks that can be inserted into a workflow. To find the ML task, just click on "Other Activities" Other activities.
The available tasks for the ML data processing are the following:

- Ml Train Elaboration
- Ml Execute Elaboration

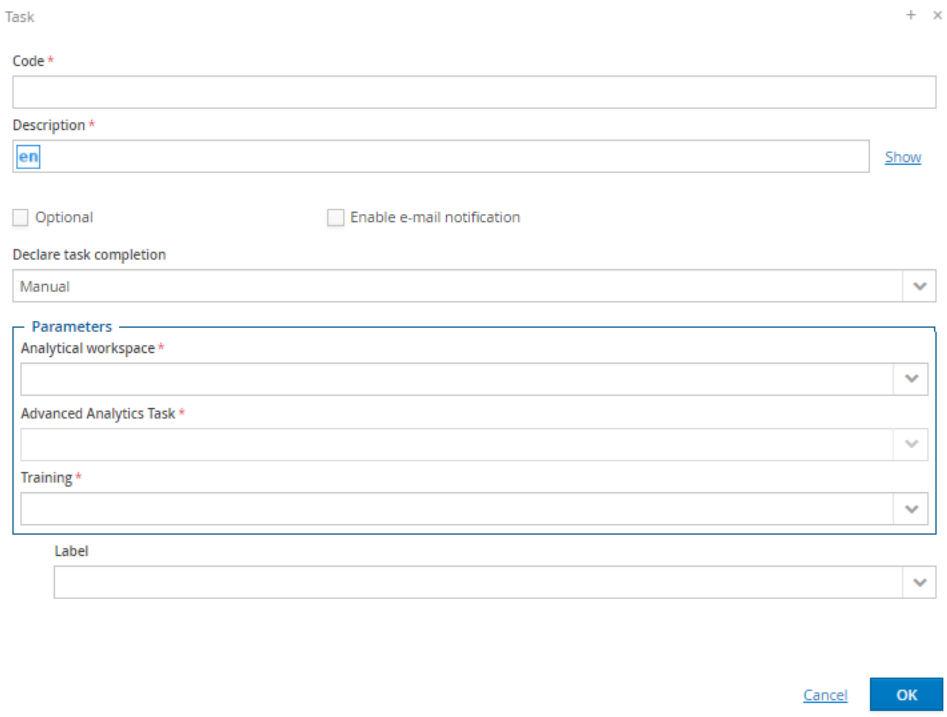
The first one allows you to run a ML inference task that runs the train of a model. Given a model, the system will ask for the Analytical Workspace, the Advanced Analytics Train previously set up in the relevant section, as shown in "Advanced Analytics" , and the Label Training which allows the user to choose the specific Training created in the "Train Label" area in Advanced Analytics.

ML Train processing

This type of task allows you to run a Train data processing in order to train a Model by using the ML engine. Once trained, the Model will be stored into a specific storage and its link will be saved in Tagetik through the previously created "Train Label", as described in Advanced Analytics; for this purpose, it is necessary first to configure the task. Once at the Entry Point , it is possible to identify the MI Train task and drag it to the selected workflow.



After that, a pop-up window in which to define the task will appear.



The pop-up window contains all the fields to fill in:

- **Code**: unique code of the task; this field is mandatory.
- **Description**: description of the MI Train task; this field is mandatory.
- **Analytical Workspace**: AW selection field.
- **Advanced Analytics Task**: Advanced Analytics of Train type configured in the Advanced Analytics Train area; this field is mandatory.
- **Training**: here the user can choose the Train Label in which the trained Model will be stored; this field is mandatory.

Moreover, selecting the **Clear output data before the data processing** checkbox deletes data from the partition of the dataset. This deletion will affect only the rows related to the Entity on which the data processing has been run, in accordance with the restrictions set out by the user rights.

After filling all fields, click "OK". The pop-up window will close and the task will be displayed in the workflow. Now, it is necessary to link the task to the workflow and then click on the "Save" button in the top right corner.

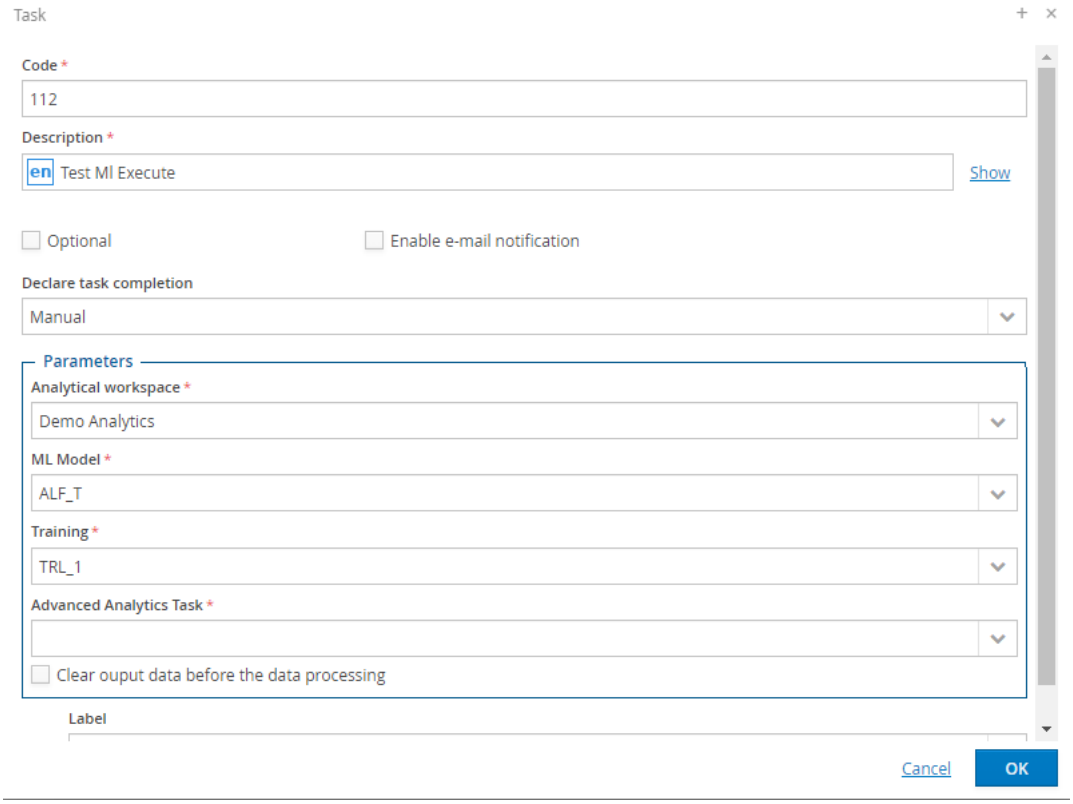


ML Execute processing

This type of task allows you to run an Execute data processing so as to run the prediction of a trained Model; to execute the prediction, the Model will use a trained Model, stored into a dedicated storage and whose link is stored in Tagetik through the "Train Label" previously used in a ML Training Task, as shows in ML Train processing; for this purpose, it is necessary first to configure the task. Once at the Entry Point , it is possible to identify the ML Execute task and drag it to the selected workflow.



After that, a pop-up window in which to define the task will appear.



The pop-up window contains all the fields to fill in:

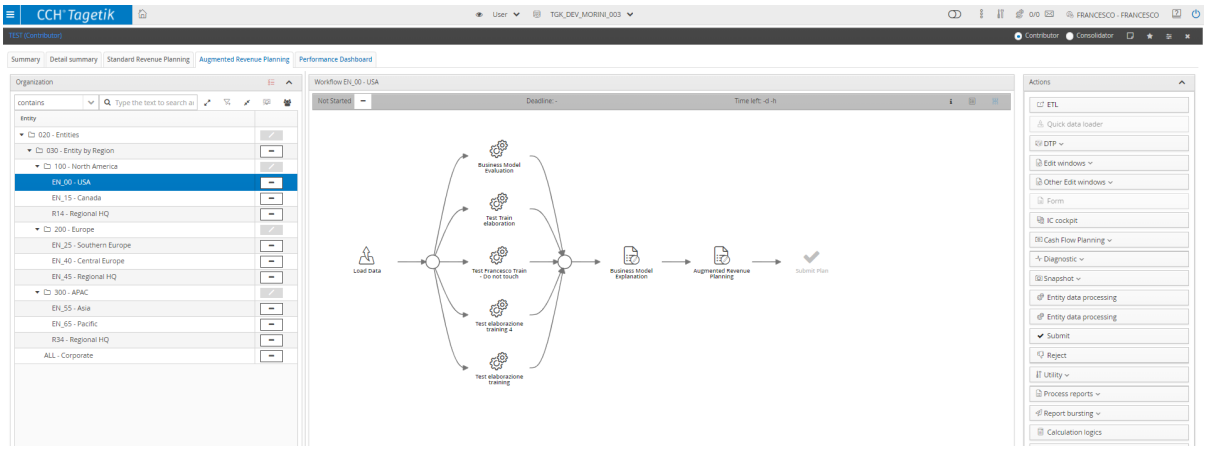
- **Code**: unique code of the task; this field is mandatory.
- **Description**: description of the MI Train task; this field is mandatory.
- **Analytical Workspace**: AW selection field.
- **MI Model**: trained Model that will be used to carry out the prediction; this field is mandatory.
- **Training**: this field allows you to choose the most suitable model through the Train Label. It is mandatory and is displayed if the chosen model has both the Train phase and the Execute phase.
- **Advanced Analytics Task**: Advanced Analytics of Execute type configured in the Advanced Analytics Execute area; this field is mandatory.

As with the ML-Training, selecting the **Clear output data before the data processing** checkbox deletes the execution Entity's data from the partition of the dataset.

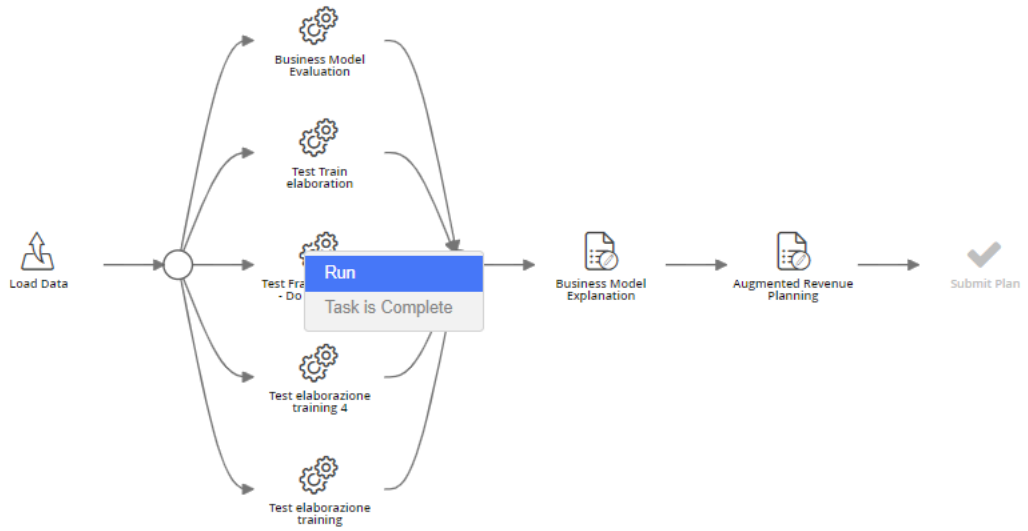
After filling all fields, click "OK". The pop-up window will close and the task will be displayed in the workflow. Now, it is necessary to link the task to the workflow and then click on the "Save" button in the top right corner.


If a Model chosen for the Execute data processing has both the TRAIN phase and the EXECUTE phase, before running an Execute task it is necessary to run a Train task using the same model chosen in the Execute data processing, otherwise the system will generate an error.

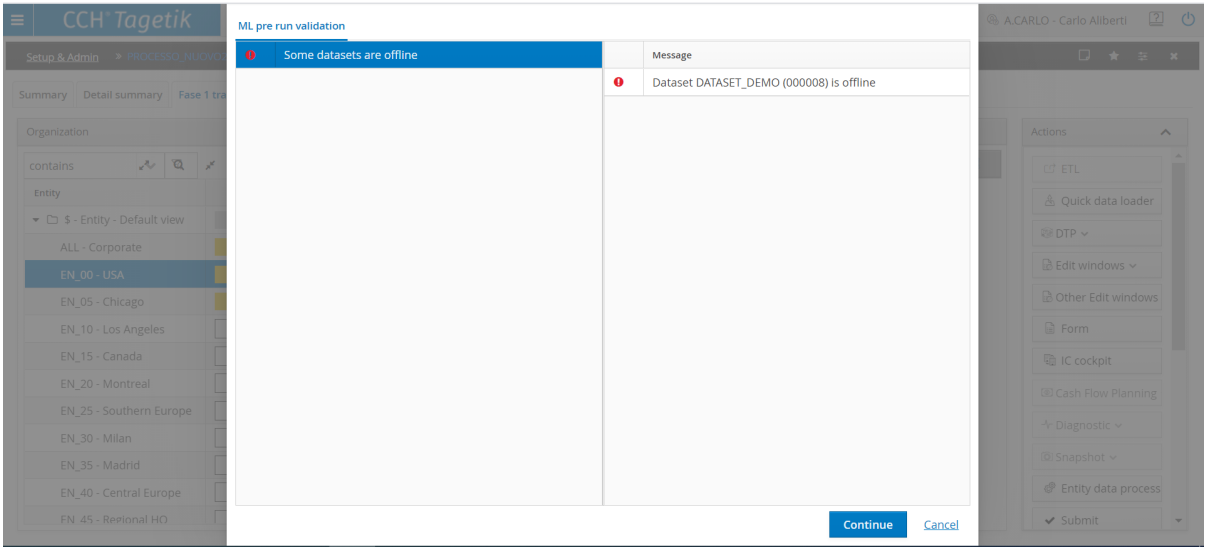
Once the Task, whether it is Execute or Train, has been created and linked to a process, it is ready to be run. To do that, it is necessary to access the User interface, click on Process from the Home Page, select the Process to which a Business Workflow containing a ML task is linked.



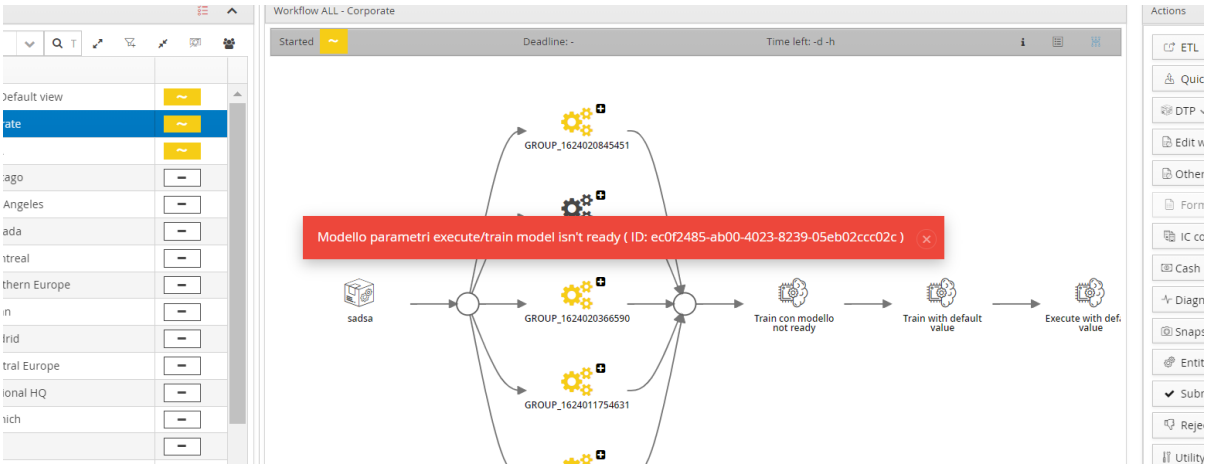
Then, after pinpointing the ML task, the user has to click and select "Run".



After clicking "Run", the system executes a *pre-run validation* to verify that the input and output datasets mapped in the AA are *online*: if even only one of them is *offline*, an error message appears and the data processing is prevented from being run.



Then the system checks the model's status. In particular, if the model's status is not 3, meaning that it is not in Ready status, an error message appears.



If the status is 3, thus it is in Ready status, the system launches the Training data processing of a Model through the Machine Learning; this will be visible in the data processing list in the top right corner by clicking on the button 1/1. The list contains also a row with "Code": OWS_ML_TRAIN_EXECUTION.

The data processing will write in output in the partition only the information related to the Entity on which the data processing has been run.

The time the data processing takes varies depending on the complexity of the uploaded Model. At the end, a pop-up window will tell the user the data processing is completed.

To make sure that the data processing has correctly written the content, an Admin User has to open the Audit and check that two rows are included in the list:

- One whose "Description" is: ML_TRAINING; this represents the recording of the reference of the trained Model in the selected Train Label.
- One whose "Description" is: ML Train Elaboration; this represents the recording of the Machine Learning's outcome into the Dataset.

When running an Execute task, the system runs a prediction data processing of a chosen Model; this will be visible in the data processing list in the top right corner by clicking on the button 1/1. The list contains

also a row with code: OWS_ML_EXECUTE_EXECUTION. The time the data processing takes varies depending on the complexity of the uploaded Model.

At the end, a pop-up window will tell the user the data processing is completed.

To make sure that the data processing has correctly written the content, an Admin User has to open the Audit and check that two rows are included in the list:

- One whose "Description" is: ML Execute Elaboration; this represents the recording of the Machine Learning's outcome into the Dataset.

When data processing starts and before it ends, it is possible to display the modelId and the correlationId in the Audit's Free Parameters section.

The Audit window displays the CorrelationID, the ElaborationID, the number of rows and allows you to make changes for each partition.

If the data processing is successfully completed but does not produce any output, the Audit will display a warning message.

ML data processes, both Train and Execute, can be configured via JOB and then scheduled.

ML task in grouping

It is possible to join two or more tasks in a group.

If a group contains a Machine Learning task linked to a model that does not contain any parameter in the descriptor.json, the entire group of tasks will be run as normal.

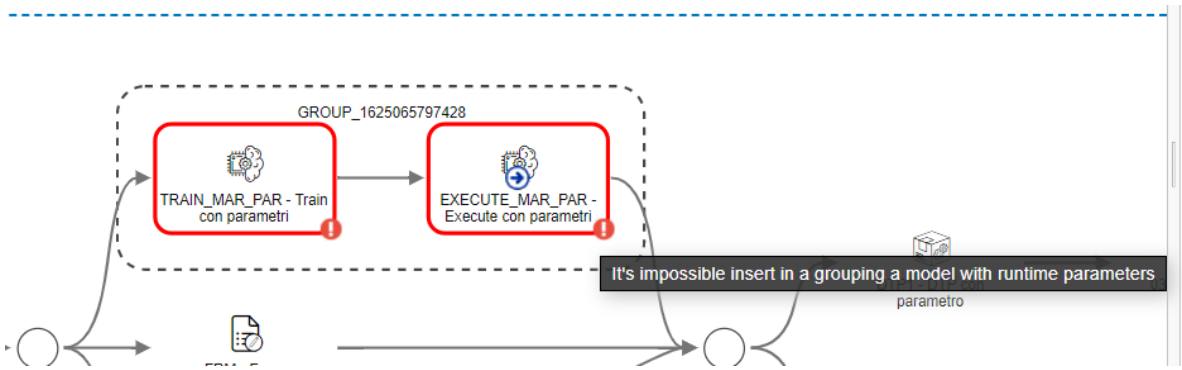
Otherwise, if a group contains a Machine Learning task linked to a model containing parameters in the descriptor.json, an error message will appear:



Any task in the group which is not a MI task with parameters will be run and processed as normal.

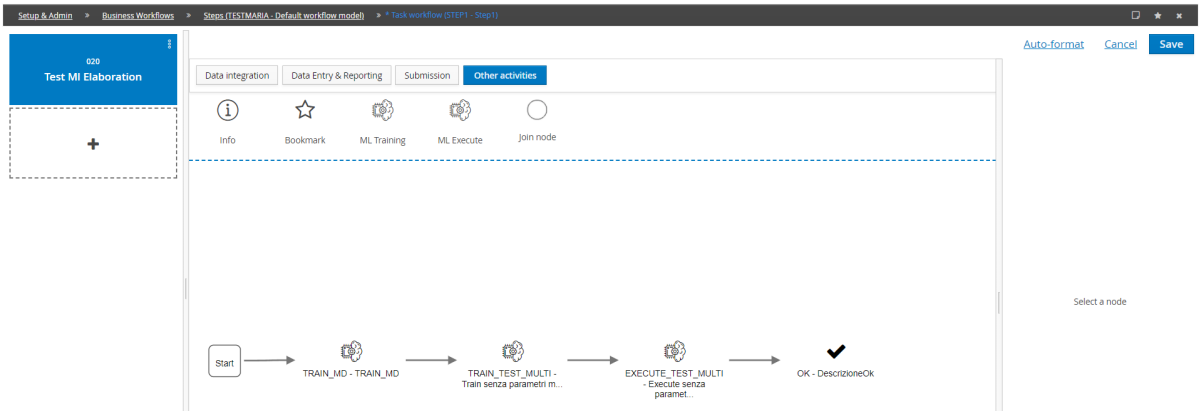
ATTENTION: The admin can no longer create groups with ML tasks linked to models with parameters in the decriptor.json.

In particular, when the admin selects two ML tasks with parameters, creates the group and clicks on Save, a red exclamation mark ! appears next to each task. When you select such exclamation mark, an error message appears and the creation of the group cannot the saved.



ML task in multidimensional process (beta version)

You can add ML tasks to a multidimensional business workflow.

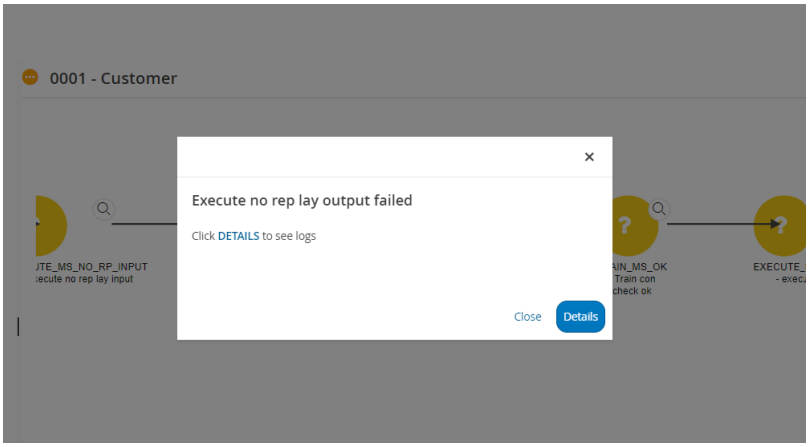


Tasks can be defined with advanced analytics linked to models either with parameters required at runtime or without parameters.

When launching a data processing, two check types are run:

- The input/output dataset cannot be partitioned by Business Cycle & Entities
- The type of the input/output dataset must be Reporting Layer.

If either of these is the case, the data processing fails and a message is shown:



By clicking on the “Details” button, you can display the details of the data processing explaining why the data processing failed.

Import Export Advanced Analytics configuration

In the **Import/Export** tool of an Analytical Workspace it is now possible to import and export the *Advanced Analytics* configurations.

The Import/Export operations include also the uploaded models and the information about the train labels and the trained models.

This option is available only if the *Machine Learning* functionality is enabled in CCH Tagetik; this can be enabled only from the Cloud service.

Entry point

The Import/Export operations can be run only by an *Admin* user, and only after having correctly set up an Advanced Analytics and its mapping, and after having uploaded at least one model and having run a training task.
The Import/Export functionality, included in the AW Import/Export tool, can be found on the Analytical Workspace view page.

Export

To export an Analytical Workspace that includes Machine Learning, the user can follow one of the following options:

- Export configuration only (models, advanced analytics train, advanced analytics execute, mapping, parameters)
- Exporting trained models

***IMPORTANT**: If only the configuration is exported, in the target environment where the Analytical Workspace will be imported, it will not be possible to run an execute without first running the train. Otherwise, if the exported Analytical Workspaces contains the trained model, in the target environment where the AW will be imported, it will be possible to run the execute on the same trained model.*

Export procedure

To export the Analytical Workspace, after navigating to the Analytical Workspaces page, you need to click the **Export** link located at the top right

A window will appear allowing the user to select the Analytical Workspace to export and to choose the **'Export Advanced Analytics Configuration'** option After checking this box, a new one 'Export trained models' is shown, which is already checked by default. If the user clicks on Run everything is exported; if the user unchecks the box 'Export trained models' only the configuration is exported. At the end of the export operation, the exported files are displayed.

IMPORTANT: It is possible to choose to export only the configuration, but it makes no sense to allow exporting only the trained models without the configuration. For this reason, if the user unchecks the “Export Advanced Analytics Configuration” box, the “Export trained models” box will automatically be hidden.

Import

To perform an import operation, it is necessary to reach to the Analytical Workspace page and press the **Import** button: the system will open a window for the selection of the file to be imported.

The standard import configuration is the one from file system: the "Select files to import" button allows you to upload a previously exported file. Once loaded, press **Next** and a new window will appear. Here you can press 

to start the import.
Once the operation is complete, the system shows the result of the import in a pop-up window.

When files containing train labels with context filtering are imported, a check is performed on the process, scenario, period, step, entity or analytical dimension. In particular, if the process, scenario, period, step, entity or analytical dimension linked to the train labels by context filter are not present in the new

environment, the related record will not be imported. At the end of the import operation, it is possible to see in the audit which records were not important and for what reason.

IMPORTANT: The import of Advanced Analystics configurations is only performed for non-existing models; models already on the platform are not imported and are therefore discarded. The same happens with the information concerning Train Labels and trained models.

IMPORTANT: Given an Analytical Workspace already present in the environment. in which advanced anlaytics train and/or advanced analytics execute are configured, when an import of the same analytical workspace is performed that does not contain the advanced analytics configurations within it (export performed without the 'Export Advanced Analytics configuration' option), the analytical workspace is overwritten and, consequently advanced anlaytics train and/or advanced analtytics execute will no longer be present in the analytical workspace.

If the Analytical Workspace to be imported contains installed models, some checks are made:

- The Analytical Workspace contains an installed model that in the target environment is not in the list of installable models.
- The Analytical Workspace contains an installed model whose version does not match the model in the target environment
- The Analytical Workspace contains an installed model and the user performing the import does not have the rights to install it in the target environment.

In all these cases, the import ends with a warning, and it will be possible to trace which model was not imported via the audit.

Machine Learning Update Analytical Workspace

Introduction

The **Update Analytical Workspace** feature allows a target Analytical Workspace to be updated from a source Analytical Workspace. This feature can also be used to update Machine Learning configurations (models, advanced analytics train, advanced analytics execute, parameters, and mapping)

Advanced Analytics Management

Below is the behaviour of Advanced Analytics when updating the Analytical Workspace. The result varies depending on the presence of the elements in the source and target workspace.

If advanced analytics...	... then
are present in both the source and target Analytical Workspace that share the same code	are updated in the target Analytical Workspace
are only present in the source workspace	are added to the target workspace
are only present in the target workspace	are not modified.

Update from exported Analytical Workspace

The **Update from exported Analytical Workspace** feature allows an Analytical Workspace to be updated from files exported from another environment.

Procedure

In order to run **Update from exported Analytical Workspace**, the following procedure must be followed:

1. click on the **Action** button of the Analytical Workspace you wish to update
2. select Update from exported Analytical Workspace
3. pick the files to be imported
4. click on **Run**.

***IMPORTANT**: The update cannot be performed if the source workspace containsthe the following types of Machine Learning configurations:*

- manually uploaded models: the Run button is not enabled and the system displays the message: "The model in the imported files is an uploaded model
- The source workspace contains a model with a version that is newer than the one supported in the target environment

ML model version management

Based on the version difference between source and target, the system behaves as follows:

If the model version in the Source...	... then the model
is older than the target	is not updated and the audit will display the message: 'The target model version is greater than the source model version'
is the same as the target	is not updated
is newer than the target and the target environment allows the use of that version	is updated
is newer than the target, the version step is a major and the target environment allows that version to be used	is updated and the audit will display the message: 'The change is a major version, so you might miss the trains'.

***Note**: If the model to be updated is in a **BLOCKED BY USER** status, then the status will remain the same even after updating the Analytical Workspace.*

Best Practices

- Generally, the training of a model is a time-consuming phase, therefore it is advisable to carry it out long before the execute phase.
- It is advisable to write the output on a type different from Reporting Layer as the data processing is provided with a deletion option for the previous execution that deletes the entire output. If the execute data processing is run by several users (e.g. each user for their own cost centre) the last execution will prevail. Whereas, if you write on the Reporting Layer, thanks to the functionalities provided by DTP and Data Source Mapping, you can select a specific part of the prediction.
- It is advisable not to export those Analytical Workspaces that use more than 5 models.

Introduction

This manual provides a complete description of the Deep Learning for Anomaly, Forecast, and Analysis model process within the CCH Tagetik Intelligent Platform. It includes both a functional introduction to the Deep Learning for Anomaly, Forecast, and Analysis model - its purpose, benefits, and typical use cases - and the detailed technical documentation needed to operate it.

Deep Learning for Anomaly, Forecast, and Analysis Overview

Deep Learning is becoming a foundational element of modern financial analytics. As organizations evolve toward more automated and data-driven processes, financial models must deliver fast, accurate, and scalable results. The Deep Learning for Anomaly, Forecast, and Analysis model addresses this need by adapting naturally to real-world data, whether large and complex or characterized by short, uneven historical series.

This model is more than a single analytical component: it is part of the neural core of the CCH Tagetik Intelligent Platform. Designed for speed, accuracy, explainability, and governance, it provides a reliable foundation for daily operations and for the next generation of intelligent automation and agentic AI capabilities.

When and Why to Model

The Deep Learning for Anomaly, Forecast, and Analysis model is particularly effective when data structures, business drivers, and KPIs interact in complex ways. It learns from multiple indicators simultaneously, enabling a unified approach to forecasting, anomaly detection, and driver analysis.

You should consider using this model when:

- Rapid execution times are required
- Data sets include large combinations of rows and dimensions
- KPIs depend on several interacting drivers
- Historical data is short, irregular, or partially incomplete
- Forecasting and anomaly detection need to follow a single logic
- Simulations and driver analysis require consistent, repeatable outputs

These are indicators, not prerequisites. Even one of these conditions suggests that Deep Learning for Anomaly, Forecast, and Analysis can provide stronger, more consistent analytical support.

Through a single training process, the model delivers predictive forecasts, anomaly detection, driver impact analysis, and validation metrics within one coherent framework, ensuring analytical consistency across Finance.

Model Design

The Deep Learning for Anomaly, Forecast, and Analysis model is a native multi-target neural network that supports multiple financial analytics needs within a single framework. Through a single training process, it enables:

- Predictive forecasting for multiple KPIs
- Automatic anomaly detection based on the same analytical foundation
- Driver impact analysis, showing how drivers influence each KPI
- Model validation metrics for evaluating reliability and performance
- Explainable, governance-ready outputs for transparent decision support

By integrating these capabilities, the model minimizes analytical fragmentation and maintains consistent logic across planning, analysis, and monitoring activities. Controllers, planners, and analysts can depend on a single, unified engine for aligned simulations, scenarios, and data-driven evaluations.

Appendix A. Input and Output Data Sets

There are two input data sets: one for the Training phase and one for the Execution phase.

There are seven output data sets: four for the Training phase and three for the Execution phase.

It is recommended to set each numerical field in the following output data sets to five decimal places to avoid errors caused by large numerical values.

Starting with Tagetik version SP31, it is possible to use data sources as input for both the Training and Execution phases.

A.1 Input Training Data Set

This data set contains historical data:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- REF_DATE_1: The first field that identifies the time of the series (text)
- REF_DATE_2: The second field that identifies the time of the series (text)
- ...
- AMOUNT: The amount for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE: The name of the variable
- TARGET_WEIGHTS: The weight (from 0 to 1) used to identify target variables and to associate a weight during validation for the datapoint

The REF_DATE and DIMENSION columns may have a variable number of columns depending on the context.

For example:

OID	DIMENSIONS_0	TARGET_WEIGHTS	REF_DATE_1	REF_DATE_2	AMOUNT	MEASURE
ab23c617	EN_01	1	2025_ACTUAL	01	7530	VOLUME
ab23c618	EN_01	1	2025_ACTUAL	02	8593	VOLUME
ab23c640	EN_01	1	2025_ACTUAL	03	9850	VOLUME
...	...	...	...	...	...	...

A.2 Input Execution Data set

This data set contains data to use for prediction:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- REF_DATE_1: The first field that identifies the time of the series (text)
- REF_DATE_2: The second field that identifies the time of the series (text)
- ...
- AMOUNT: The amount for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series

- ....
- MEASURE: The name of the variable

The REF_DATE and DIMENSION columns must be defined according to the training step.

For example:

OID	DIMENSIONS_0	TARGET_WEIGHTS	REF_DATE_1	REF_DATE_2	AMOUNT	MEASURE
ab23c617	EN_01	1	2026_ACTUAL	01	8510	VOLUME
ab23c618	EN_01	1	2026_ACTUAL	02	9892	VOLUME
ab23c640	EN_01	1	2026_ACTUAL	03	9979	VOLUME
...	...	...	...	...	...	...

A.3 Output Training Data Set for Metrics

This data set contains the metrics of the model:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- AMOUNT: The metric value for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE: The name of the variable
- METRIC: The metric (text)

For example:

OID	DIMENSIONS_0	AMOUNT	METRIC	MEASURE
ab23c617	EN_01	0.045	MAPE	VOLUME
ab23c700	EN_01	0.140	R2	SALES
...	...	...	...	...

A.4 Output Training Data Set for Fit

This data set contains the fit of the model:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- REF_DATE_1: The first field that identifies the time of the series (text)
- REF_DATE_2: The second field that identifies the time of the series (text)
- ...
- FIT: The fit amount for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE: The name of the variable

For example:

OID	DIMENSIONS_0	REF_DATE_1	REF_DATE_2	FIT	MEASURE
ab23c617	EN_01	2025_ACTUAL	01	8610	VOLUME
ab23c618	EN_01	2025_ACTUAL	02	9800	VOLUME
ab23c640	EN_01	2025_ACTUAL	03	9974	VOLUME
...	...	...	...	...	...

A.5 Output Training Data Set for Importance

This data set contains the forecast of the model:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- IMPORTANCE: The importance value of an external feature for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE_TARGET: The name of the target variable
- VARIABLE: The name of the external feature (text)
- LAG: The lag of the external feature (text)**

** If the variable (external feature) is the seasonality feature, the lag column contains the seasonality information for that variable. If the variable is a dimension feature, the lag column remains blank.

For example:

OID	DIMENSIONS_0	VARIABLE	LAG	IMPORTANCE	MEASURE_TARGET
ab23c617	EN_01	PRICE	-1	0.30	VOLUME
ab23c618	EN_01	PRICE_SEASONALITY	6	0.25	VOLUME
ab23c640	EN_01	DIMENSIONS_0		0.45	VOLUME

A.6 Output Training Data Set for Contribution

This data set contains the contributions of the exogenous variable of the model:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- REF_DATE_1: The first field that identifies the time of the series (text)
- REF_DATE_2: The second field that identifies the time of the series (text)
- ...
- CONTRIBUTION: The contribution value of an external feature for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE_TARGET: The name of the target variable
- VARIABLE: The name of the external feature (text)
- LAG: The lag of the external feature (text)**

** If the variable (external feature) is the seasonality feature, the lag column contains the seasonality information for that variable. If the variable is a dimension feature, the lag column remains blank.

For example:

OID	DIMENSIONS_0	VARIABLE	LAG	CONTRIBUTION	REF_DATE_0	REF_DATE_1	MEASURE_TARGET
ab23c617	EN_01	PRICE	-1	200	2025_ACTUAL	01	VOLUME
ab23c618	EN_01	PRICE_SEASONALITY	6	60	2025_ACTUAL	01	VOLUME
ab23c640	EN_01	DIMENSIONS_0		15	2025_ACTUAL	01	VOLUME
...	...	...	...	...			...

A.7 Output Execution Data Set for Forecast

This data set contains the forecast of the model:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- REF_DATE_1: The first field that identifies the time of the series (text)
- REF_DATE_2: The second field that identifies the time of the series (text)
- ...
- FORECAST: The forecasted amount for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE: The name of the variable
- LOWER: The lower bound of the confidence interval
- UPPER: The upper bound of the confidence interval
- IS_OUTLIER_PREDICTION: The prediction of outlier/inlier
- OUTLIER_PROBABILITY: The probability of being an outlier

For example:

OID	DIMENSIONS_0	REF_DATE_1	REF_DATE_2	FORECAST	MEASURE	UPPER	LOWER	IS OUTLIER PREDICTION	OUTLIER PROBABILITY
ab23c617	EN_01	2025_ACTUAL	01	8610	VOLUME	1000	7220	1	0.7
ab23c618	EN_01	2025_ACTUAL	02	9800	VOLUME	1200	7600	0	0.1
ab23c640	EN_01	2025_ACTUAL	03	9974	VOLUME	1300	6.948	0	0.2
...	...	...	...	...	..		...	...	

A.8 Output Execution Data Set for Importance

This data set contains the forecast of the model:

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- IMPORTANCE: The importance value of an external feature for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE_TARGET: The name of the target variable
- VARIABLE: The name of the external feature (text)
- LAG: The lag of the external feature (text)**

** If the variable (external feature) is the seasonality feature, the lag column contains the seasonality information for that variable. If the variable is a dimension feature, the lag column remains blank.

For example:

OID	DIMENSIONS_0	VARIABLE	LAG	IMPORTANCE	MEASURE_TARGET
ab23c617	EN_01	PRICE	-1	0.50	VOLUME
ab23c618	EN_01	PRICE_SEASONALITY	6	0.15	VOLUME
ab23c640	EN_01	DIMENSIONS_0		0.35	VOLUME
...	...	...	...	...	...

A.9 Output Execution Data Set for Contribution

This data set contains the contribution of each variable and date in the prediction.

- OID: This is the first field, and it identifies the row ID (it does not need to be mapped; it is just present in the data)
- REF_DATE_1: The first field that identifies the time of the series (text)
- REF_DATE_2: The second field that identifies the time of the series (text)
- ...
- CONTRIBUTION: The contribution value of an external feature for each series (number)
- DIMENSIONS_0: The first field that identifies the time series
- DIMENSIONS_1: The second field that identifies the time series
- ....
- MEASURE_TARGET: The name of the target variable
- VARIABLE: The name of the external feature (text)
- LAG: The lag of the external feature (text)**

** If the variable (external feature) is the seasonality feature, the lag column contains the seasonality information for that variable. If the variable is a dimension feature, the lag column remains blank.

For example:

OID	REF_DATE_0	REF_DATE_1	DIMENSIONS_0	VARIABLE	LAG	CONTRIBUTION	MEASURE_TARGET
ab23c617	2025_ACTUAL	01	EN_01	PRICE	-1	100	VOLUME
ab23c618	2025_ACTUAL	01	EN_01	PRICE_SEASONALITY	6	50	VOLUME
ab23c640	2025_ACTUAL	01	EN_01	DIMENSIONS_0		25	VOLUME
...		...		...	...	...	...

Appendix B. Dimensions and Reference Date

DIMENSIONS LIST: If you need multiple dimensions to identify the time series key (e.g., ENTITY, CHANNEL, PRODUCT), you can map the fields directly in the configuration window.

REF_DATE: If you need multiple dimensions to identify the time ID for the historical series (e.g., SCENARIO, PERIOD), you can map the fields directly in the configuration window.

For example:

Appendix C. Parameters

The parameters to set for the Training phase include the following:

- LOG LEVEL:
- DEBUG: all information logged
- INFO: all warnings, errors, critical errors, and general information logged without debugging information
- WARNING: all warnings, errors, and critical errors are logged

- ERROR: all errors and critical errors are logged
- CRITICAL: only critical errors are logged

Logging messages that are less severe than the specified level are ignored, while messages with a severity level or higher are logged.

- FEATURES_MAX_LAG: Maximum lag applicable
- USE_TARGET_FEATURE_SEASONALITY: whether to create the seasonal feature of the target or not.
- USE_EXOGENOUS_FEATURES_SEASONALITY: whether to create or omit the seasonality features of the exogenous variables.
- LAG_TARGET_FEATURES: whether to generate lagged features of the target features.
- LAG_EXOGENOUS_FEATURES: whether to create lagged features of exogenous variables or not.
- GET_CONTRIBUTIONS: used to retrieve driver contributions as output.
- GET_IMPORTANCES: to obtain the driver's importance as output.
- SCALE_FEATURES: to scale the input features before training the neural network.
- FEATURES_SELECTION_THRESHOLD: (from 0 to 1) the threshold used to determine whether a feature should be excluded based on its importance.
- INCLUDE_OTHERS_TARGETS_IMPACT: to include the lags of other targets as features for a target with the same dimensionality.
- Use the lagged exogenous features: Y (Yes) or N (No) to create the lagged version of the exogenous features. The default is Y (Yes), but you can set it to N (No) to reduce processing time.
- Feature max lag: the maximum lag value that can be applied to the selected variables (target or exogenous). The default is AUTO, which corresponds to the seasonal value.
- Use the lagged target feature: Y (Yes) or N (No) to create the lagged version of the target features. The default is Y (Yes), but you can set it to N (No) to reduce processing time.

The parameters to set for the Execution phase include:

- LOG LEVEL:
- DEBUG: all information logged
- INFO: all warnings, errors, critical errors, and general information logged without debugging information
- WARNING: all warnings, errors, and critical errors are logged
- ERROR: all errors and critical errors are logged
- CRITICAL: only critical errors are logged

Logging messages that are less severe than the specified level are ignored, while messages with a severity level or higher are logged.

- CONFIDENCE_LEVEL: the confidence level used to calculate the confidence interval.
- GET_CONTRIBUTIONS: used to retrieve driver contributions as output.
- GET_IMPORTANCES: to obtain the driver's importance as output.
- EXECUTE_ANOMALY: to perform anomaly detection on the target having actual values as input execute.

Deep Learning for Anomaly, Forecast, and Analysis Model

The Deep Learning for Anomaly, Forecast, and Analysis model can be divided into three steps: Data Preparation, Training, and Execution.

The goal is to prepare data for the forecasting task (Data Preparation), then fit the model with minimal error (Training), and predict the target variable values (Execution).

This model is designed to train only one Deep Learning architecture once; it performs a single training pass over all historical series. That same model then provides predictions for all dimensions and targets in the data.

The solution relies on the OData API Service, and the user setting it up must have the necessary permissions.

Data Preparation

The goal of this step is to prepare data for the forecasting task.

- Duplicated values are managed by summing the corresponding values (aggregation of the rows with the same dimensionality ID and time ID).
- Exogenous features that are not associated with any target are excluded: they must have the same dimensionality as the target you aim to influence.
- Numerical data of the exogenous variables are scaled to a mean of 0 and a variance of 1.
- The seasonality for each feature is calculated.
- Optionally, seasonality features for the target and exogenous variables can be created using a rolling mean at seasonal intervals. There is a parameter that controls whether to generate seasonal variables for the target and exogenous features.
- For each feature, the script creates lagged features by shifting the original data from 1 up to the smaller of the maximum seasonality of that feature (across different dimensions) and MAX_LAG, if it is set. After generating the lagged features, some missing values will appear at the start of the data set. All of these must be removed to ensure proper model training. As a result, the data set will lose several rows equal to the maximum lag across all features, multiplied by the number of dimensions.

Therefore, to retain as much data as possible, we suggest keeping MAX_LAG relatively low.

- If the flag for scaling is set to Y, during the pre-processing step, we apply the standard scaler to the external features and targets.

Example:

The features in the input training data set are:

COVID_RATE

MARKETING

While VOLUME is the target variable.

The script calculates the seasonality of the target variable, which is 2, and the seasonality of each feature.

COVID_RATE - 3

MARKETING - 2

Given a MAX_LAG of 5, the script calculates the shift up to the minimum between 5 and the maximum seasonality of each feature (minus 1):

COVID_RATE	COVID_RATE-1	COVID_RATE-2	MARKETING	MARKETING-1
0.8			95	
0.7	0.8		100	95
0.75	0.7	0.8	90	100
0.9	0.75	0.75	110	90
...	...	...	...	...

As you can see, this step produces some missing values in the data set, so the first two rows will be deleted

COVID_RATE	COVID_RATE-1	COVID_RATE-2	MARKETING	MARKETING-1
0.75	0.7	0.8	90	100
0.9	0.75	0.75	110	90
...	...	...	...	...

Training

The goal of this step is to develop a custom Multi-Task Deep Neural Network architecture.

The last 20% of the data is reserved for validation, allowing the model's error to be evaluated by comparing actual and forecasted values.

The model is optimized by minimizing the Mean Absolute Percentage Error (MAPE).

The contribution of each driver is calculated at each time point for every target. Then, the importances are determined as the average contribution per target and normalized so that they sum to 1 for each target. If the contributions are set to N, the importances are not derived from these contributions but are instead calculated from the final gradients associated with each input variable.

This phase creates four data sets:

- Fit data set – containing the fitted values
- Importance data set – summarizing driver importances
- Contribution data set – detailing contributions over time
- Metrics data set – reporting performance metrics

In the metrics data set, for each target, four metrics (R^2, MSE, MAE, and MAPE) are provided, calculated on the validation data set. Additionally, the standard deviation of the residuals is included.

Execution

The goal of this step is to predict the target variable values using the trained model.

If the forecasting horizon extends beyond one step (meaning more than one point to predict), a rolling forecast strategy is used to generate predictions for the entire required period. In this method, previously forecasted values serve as inputs for the subsequent predictions.

This phase produces three data sets:

- Forecast data set – includes predicted values with their confidence intervals
- Importance data set – summarizes driver importances
- Contribution data set – shows contributions over time

If anomaly detection is performed on the actual target values provided as input, the forecast data set includes an additional column showing the outlier/inlier classification and the associated outlier probability.